Cosmology

Leonard Susskind

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Transcript-derived notes curated by LazyingArt LLC with Video2Book

Original lectures by Leonard Susskind. Transcript-derived course notes curated by LazyingArt LLC with Video2Book.

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Chapter 1

Geometry of the Expanding Universe

These notes follow Leonard Susskind's first cosmology lecture as closely as possible in both argument and pace. The transcript, together with selected blackboard frames preserved through LazyingArt LLC and Video2Book, lets us keep the order of discovery clear: first a toy expanding universe, then Hubble's law, then the metric of space, and only then the metric of spacetime. The lecture is deliberately geometry-first. Newtonian cosmology, Einsteinian dynamics, and the full Friedmann–Robertson–Walker package are promised for later. Here we are interested in a simpler question: what does it mean, geometrically, to say that the universe expands?

1.1 Geometry First, Dynamics Later

Susskind begins by declining to tell the whole history of cosmology. We are to get straight to the point, and the point is the expanding universe and its geometry. That opening is not merely rhetorical. It tells us why this chapter begins with idealized lines, grids, and metrics, and not with Einstein's equations.

The roadmap is brief but important. We will study the geometry of expanding universes first and leave the physics of expansion for later. Newtonian $F = ma$ cosmology will come next, and only after that will geometry and dynamics be fused into the full FRW description. The present lecture therefore asks us to be disciplined: for now we describe the universe; we do not yet ask what drives the description.

1.2 Comoving Coordinates and the One-Dimensional Expanding Line

We begin with the simplest possible space: one-dimensional space, *space, not spacetime.* Take an infinite line, place equally spaced galaxies on it, and label the galaxies by a coordinate x. The first conceptual point is that the coordinate is not yet a measured distance. It is a tag attached to a galaxy, and it moves with the galaxy if the universe expands or contracts. One may think of x as the name of the galaxy.

Suppose the distance from the galaxy called $x = 0$ to the galaxy called $x = 1$ is some measured quantity a. Then the distance between any two galaxies is the coordinate difference multiplied by

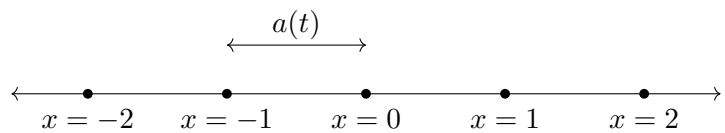


Figure 1.1: A one-dimensional universe of equally spaced galaxies. The labels $x = -2, -1, 0, 1, 2$ are comoving coordinates; the neighboring physical separation is the scale factor $a(t)$.

this common spacing:

$$d = a\,\Delta x. \tag{1.1}$$

This is the first mathematical move that matters. The coordinate difference Δx is not itself the physical distance. It only records which two galaxies we have chosen. The scale factor a converts a comoving label difference into a true geometric separation.

Now we imagine the line as an elastic thread. Two giants pull on its ends and stretch it or let it contract. The labels remain attached to the galaxies, but the neighboring physical spacing changes. Thus a becomes time-dependent:

$$d(t) = a(t)\,\Delta x. \tag{1.2}$$

At this stage homogeneity means simply that neighboring galaxies are equally spaced everywhere. The common spacing may still vary with time.

This is already enough structure to say something nontrivial. We have a universe in which points carry fixed comoving labels while the physical distance between them changes. That is the raw material from which the rest of the lecture is built.

1.3 From the Scale Factor to Hubble's Law

We now calculate the rate at which the distance between two fixed galaxies changes. The word fixed matters: we keep the same pair of galaxies, so Δx is constant in time. Differentiating $d = a\,\Delta x$ gives

$$\dot{d} = \dot{a}\,\Delta x \tag{1.3}$$

$$= \frac{\dot{a}}{a}\,a\,\Delta x \tag{1.4}$$

$$= \frac{\dot{a}}{a}\,d. \tag{1.5}$$

This is Hubble's law in its cleanest form. If we define

$$H(t) \equiv \frac{\dot{a}}{a}, \tag{1.6}$$

then the law reads

$$\dot{d} = H(t)\,d. \tag{1.7}$$

Susskind follows the standard historical language and says "Hubble constant," but he immediately pauses to correct the impression that the name gives. The quantity is constant in space, not in time. It is the same for every galaxy at a given moment, but in general it evolves with the universe. So in the notes we should hear his warning and keep in mind that $H(t)$ is really the Hubble parameter.

The physical content comes next. Galaxies farther away recede faster. But the argument does not pick out a center. Nothing in the formula privileges $x = 0$, $x = 1$, or any other point. The law concerns relative separation, and it is the same everywhere. The expanding universe described here is therefore not a blast from one distinguished place.

Only after deriving the law does the lecture turn outward toward observation. If Alice and Bob exchange light signals while moving apart, the received light is redshifted. The faster the recession, the greater the redshift. Spectral lines therefore offer a way to estimate recession speed. Distance is harder, but Hubble used rough observational surrogates such as brightness and angular size. His plot was not a perfect straight line. It was noisy, with large scatter, and his original distances were badly misestimated. But the geometric principle was right: recession speed is proportional to distance.

A second caution is briefly raised and deferred. If d is large enough, the formula predicts $\dot{d} > 1$ in units with $c = 1$. Susskind flags that as important, but leaves the resolution for later lectures.

1.3.1 Question & Answer

Question. Does $v = Hd$ mean that a given pair of galaxies must be accelerating, simply because the distance d is increasing?

Answer. No. The law says that velocity increases with distance at a given moment. It does *not* by itself say whether the velocity of one fixed pair increases with time. That depends on how $a(t)$ behaves.

If

$$a(t) = \alpha t, \tag{1.8}$$

then neighboring galaxies separate at constant speed:

$$\dot{d} = \alpha \, \Delta x. \tag{1.9}$$

The recession speed depends on the pair we choose, but it does not change with time.

If instead

$$a(t) = \beta t^2, \tag{1.10}$$

then

$$\dot{d} = 2\beta t \, \Delta x, \tag{1.11}$$

so the recession speed of a fixed pair increases with time.

And if

$$a(t) = \gamma \log t, \tag{1.12}$$

then

$$\dot{d} = \frac{\gamma}{t} \, \Delta x, \tag{1.13}$$

so the recession speed decreases with time.

Thus Hubble's law is always true as a function of distance. Whether the universe is accelerating or decelerating is a second question, encoded in the detailed time dependence of $a(t)$. Susskind uses exactly these elementary examples to separate the two issues before moving on.

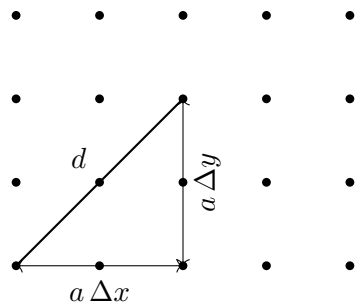


Figure 1.2: A two-dimensional comoving grid. The lecture uses such a grid only as a geometric idealization, but it makes the roles of homogeneity, isotropy, and the Pythagorean theorem transparent.

1.4 Higher Dimensions, Isotropy, and Homogeneity

At this point the lecturer says, in effect, that there is nothing especially one-dimensional about the argument. So let us widen the stage. In two dimensions we imagine a square grid of galaxies. Each galaxy carries two labels, (x, y), and these labels again move with the galaxies rather than measuring physical distance directly.

Before writing the distance formula, the lecture asks a natural question: why should the spacing along the x-direction be the same as the spacing along the y-direction? The answer is not abstract formalism; it is observation. If the galaxy distribution were systematically more crowded along one axis than another, we would see that anisotropy in the sky. On sufficiently large scales, we do not. That empirical fact is what motivates using the same scale factor in all spatial directions.

With that understood, the distance between two galaxies separated by Δx and Δy is found from the Pythagorean theorem:

$$d = a\sqrt{(\Delta x)^2 + (\Delta y)^2}. \tag{1.14}$$

And in three dimensions the extension is immediate:

$$d = a\sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}. \tag{1.15}$$

For a fixed pair of galaxies the coordinate differences are constant, so differentiation once again gives

$$\dot{d} = \frac{\dot{a}}{a}\, d. \tag{1.16}$$

The Hubble law survives unchanged.

Now the lecture names the two symmetries that have been guiding the construction. Homogeneity means sameness from place to place. Isotropy means sameness from direction to direction. Susskind is careful not to over-polish this into textbook dogma. The universe is not homogeneous on the scale of a galaxy, nor even on modest astronomical scales. It is not perfectly isotropic when we look within the Milky Way. But after averaging over many galaxies, the large-scale sky looks roughly the same in every direction, and that is the working content of the cosmological principle at this stage.

1.5 Why the Scale Factor Is a Bookkeeping Device, and Why the Metric Matters

The lecture next doubles back to a subtle point that is easy to miss if we move too fast. If $H = \dot{a}/a$ is what enters Hubble's law, does the overall numerical value of a by itself have any physical meaning? Susskind's answer is no. The scale factor is, in part, a bookkeeping device. One may redefine it by a constant multiple without changing the observable Hubble law.

If we introduce a new scale factor b by

$$b = 3a, \tag{1.17}$$

then

$$\frac{\dot{b}}{b} = \frac{3\dot{a}}{3a} = \frac{\dot{a}}{a}. \tag{1.18}$$

So the overall normalization of the scale factor is conventional. What matters physically is how it changes, not its absolute magnitude. This is the lecture's answer to the question about the "initial value" of a: one may choose units or normalization freely, but the ratio $\dot{a}/a$ does not care.

Only after that clarification does Susskind turn from kinematics to geometry proper. For neighboring points we replace Δx by dx. Then the physical distance between neighboring points is

$$ds = a\,dx, \tag{1.19}$$

and therefore

$$ds^2 = a^2\,dx^2. \tag{1.20}$$

This is now a spatial metric. In one dimension the coefficient of dx^2 is simply

$$g_{xx} = a^2. \tag{1.21}$$

So an expanding universe can be described by saying that the spatial metric itself is time-dependent.

The lecture then pushes one step further and asks for the spacetime metric. Proper time must include both the time separation and the physical spatial separation. Since the physical spatial distance is $a\,dx$, the natural one-dimensional spacetime line element becomes

$$d\tau^2 = dt^2 - a^2(t)\,dx^2. \tag{1.22}$$

For a comoving observer, $dx = 0$, so proper time reduces to ordinary time. For an object moving across the comoving grid, the spatial term contributes in the usual relativistic way.

The frame below is worth keeping precisely because it shows how the lecture makes the transition on the board: spacetime sketch on the left, cleaned line element in the middle, metric notation on the right.

In three spatial dimensions the board writing is only partly visible, so the standard clean form should be read as a cautious reconstruction supported by both the transcript and the frame:

$$d\tau^2 = dt^2 - a^2(t)\big(dx^2 + dy^2 + dz^2\big). \tag{1.23}$$

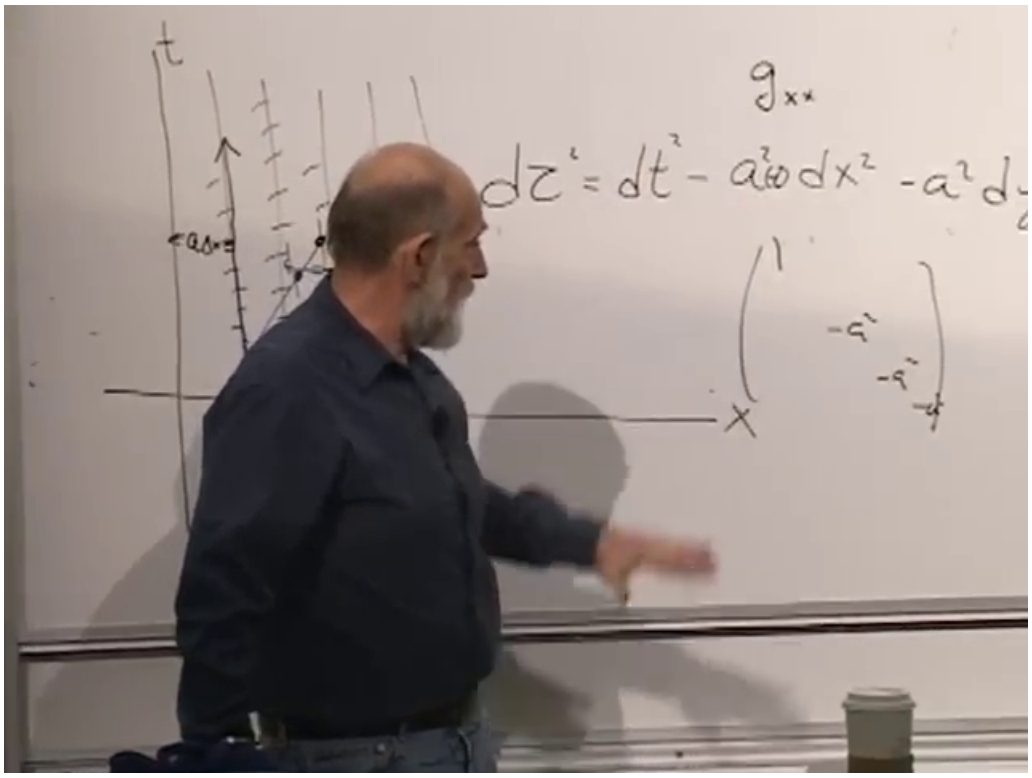
Likewise, the matrix on the right side of the board is partly occluded, but the intended diagonal form is clear enough to write it carefully as

$$g_{\mu\nu} = \mathrm{diag}\big(1, -a^2, -a^2, -a^2\big). \tag{1.24}$$

The off-diagonal terms vanish:

$$g_{tx} = g_{ty} = g_{tz} = 0, \tag{1.25}$$

which is the lecture's way of saying that in this simple ansatz there are no mixed $dt\,dx$-type terms.



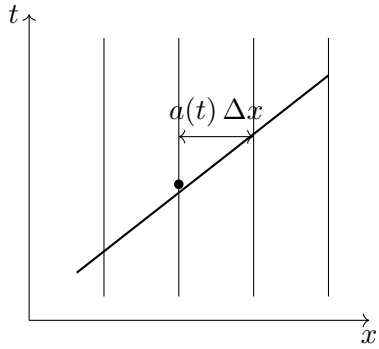


Figure 1.3: Expanding-universe spacetime sketch and metric evidence. The screenshot preserves the lecture's board layout; the redraw isolates the comoving worldlines and the fixed-time spatial separation.

1.5.1 Question & Answer

Question. Is dx a physical distance, or only a comoving label?

Answer. It is only a comoving label interval. The physical distance is $ds = a\,dx$. The clean way to think about it is exactly the way Susskind phrases it: draw a coordinate mesh through the galaxies and decree that the mesh follows them. Then Δx between two chosen galaxies stays fixed by construction, while the measured distance between them changes because the scale factor changes. One may place the units in x, or in a, or split them between the two; that is convention. What is not conventional is the product $a\,dx$, and what is especially invariant is the ratio $\dot{a}/a$.

The lecture also pauses here over another convention. The coefficient of dt^2 has been set equal to 1, and Susskind remarks that if that coefficient depended only on time, one could absorb it into a redefinition of the time variable. Similarly, he adopts units with $c = 1$. None of that changes the main point: expansion is the time dependence of the spatial components of the metric.

We are still not doing dynamics. One could now ask what energy-momentum tensor inserted into Einstein's equations would produce this metric, but the lecture explicitly postpones that. Geometry first, dynamics later.

1.6 What We Know, What We Assume, and What the Horizon Hides

With the metric in hand, Susskind makes what he calls a slightly finer distinction. Homogeneity means the metric coefficients do not depend on spatial position. Isotropy means the spatial directions are treated alike. In the diagonal metric above, the fact that the coefficients do not depend on x, y, or z is the metric statement of homogeneity. The fact that the three spatial coefficients are equal is the metric statement of isotropy.

1.6.1 Question & Answer

Question. What exactly do homogeneity and isotropy buy us, and how are they related?

Answer. They buy us an enormous simplification. Homogeneity says the metric can depend on time but not on where we are in space. Isotropy says the x-, y-, and z-directions must enter in the same way. The two ideas are not identical: a universe may be homogeneous and still have a preferred direction. But isotropy about every point forces homogeneity. That is why the lecture treats them as distinct concepts that are closely related rather than identical principles.

The lecture then becomes deliberately cautious. The universe is not exactly homogeneous at small scales, nor exactly isotropic in every local observation. A farmer's field provides the analogy: on one scale it looks homogeneous, on a smaller scale it plainly does not, and on a scale larger than the whole field it again ceases to look homogeneous. Cosmology is similar. On scales of galaxies the universe is highly structured; only after averaging over much larger scales does the rough uniformity emerge.

That caution widens into a discussion of the horizon. We do not see the whole universe. More

strongly, we do not even see all of the universe that could ever in principle affect us. Beyond the observable region we do not know whether homogeneity and isotropy continue to hold. There may be indirect clues, but the lecture insists that the horizon is an ultimate barrier to direct knowledge.

This is why the cosmological principle is both fruitful and modest. It is a powerful empirical approximation on the scales astronomy can probe, but it is not a metaphysical guarantee about arbitrarily large scales.

1.7 Closed and Bounded Universes: The Circle Model

Only after laying out the flat expanding picture does the lecture pivot to closed geometries. The order matters. We are now varying the global shape of space while preserving the local logic of comoving labels, scale factor, and Hubble flow.

The first step is to reject the wrong bounded models. A line segment has finite extent, but it comes with distinguished endpoints. A cube with walls is finite, but it violates both homogeneity and isotropy: observers near a wall see something different from observers far from a wall, and even an observer at the center sees different things in different directions. So boundedness, if the universe is to remain homogeneous and isotropic, cannot mean a box with edges.

The right idea is bounded without boundary. Susskind first gestures toward the surface of a sphere, not the interior of a ball. A two-dimensional creature confined to the surface of a featureless sphere finds every place and every direction locally the same, yet the world is finite. That is the model of boundedness he wants. He then reduces the example to one dimension, because the algebra is simpler there: life on a circle.

In one dimension isotropy means left and right are equivalent. A circle has exactly that feature. The convenient coordinate is the angle θ, measured in radians. We choose one galaxy and call it $\theta = 0$, then label every other point by an angular coordinate. A point just to the "left" of the origin may be described by a small negative angle or, equivalently, by an angle just short of 2π. That periodicity is part of the geometry and not a side remark.

There is also a brief notational wobble in the lecture that is worth preserving in cleaned form. Susskind momentarily hesitates over whether to call the whole circumference the scale factor or to write it as 2π times a radius-like quantity. The cleaner convention, and the one we keep, is

$$L = 2\pi a, \tag{1.26}$$

where L is the total distance around the circle and a is the radius-like parameter of the external plotting circle.

Then, for a small angular separation $\Delta\theta$, the physical arc length is

$$d = a\,\Delta\theta. \tag{1.27}$$

Passing to infinitesimals gives the local metric

$$ds = a\,d\theta, \tag{1.28}$$

and therefore

$$ds^2 = a^2\,d\theta^2. \tag{1.29}$$

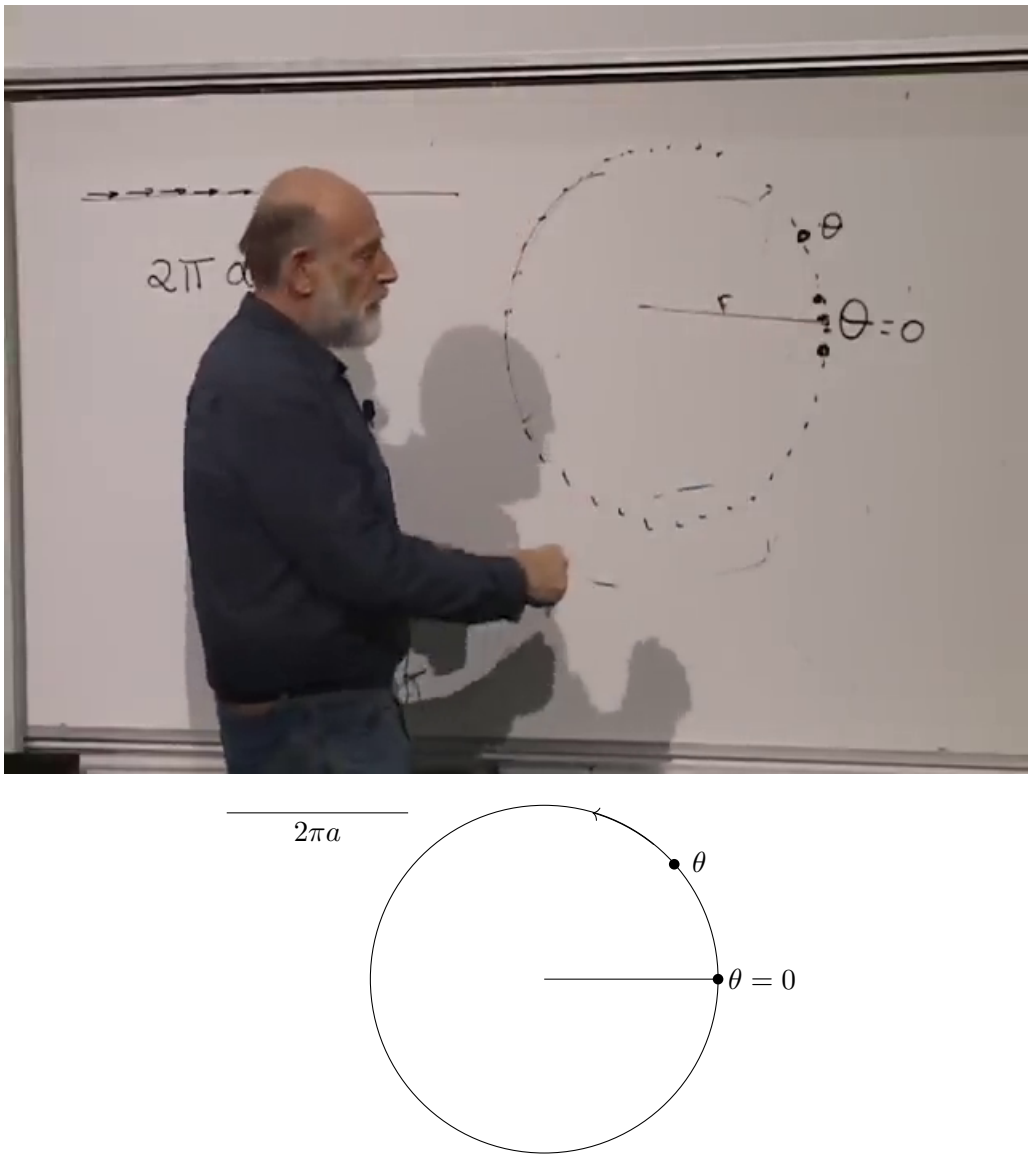


Figure 1.4: The circle model of a closed one-dimensional universe. The original frame is retained because the angular labels and the separate $2\pi a$ segment are part of the lecture's geometric reasoning, not mere ornament.

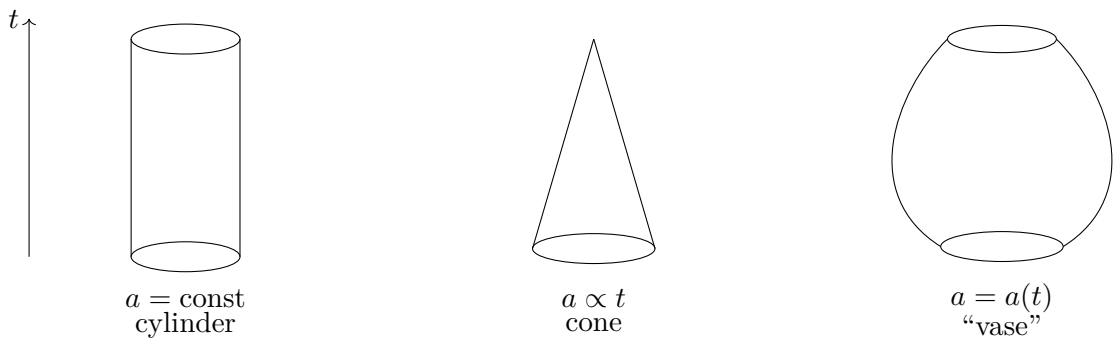


Figure 1.5: Explanatory redraw of the circle-universe spacetime for three choices of scale factor. These are external visualization aids that follow the lecture's narrative; they do not replace the intrinsic metric description.

This last equation is not directly visible in the captured frame, but it is the canonical cleaned form supported by the lecture's derivation.

Once the metric has been written, the analogy with the straight-line universe is immediate. If a depends on time,

$$d(t) = a(t)\,\Delta\theta, \tag{1.30}$$

and therefore

$$\dot{d} = \dot{a}\,\Delta\theta \tag{1.31}$$

$$= \frac{\dot{a}}{a}\,d. \tag{1.32}$$

So Hubble's law survives in the closed universe too, at least for separations small enough that we do not worry about the ambiguity of "the other way around" the circle.

The spacetime metric of the time-dependent circle model is then

$$d\tau^2 = dt^2 - a^2(t)\,d\theta^2. \tag{1.33}$$

The lecture visualizes this spacetime by stacking circles along a vertical time axis. If a is constant, the stack is a cylinder. If $a \propto t$, it is a cone. If a grows and shrinks in some more complicated way, the surface becomes vase-like.

1.7.1 Question & Answer

Question. Where did the Big Bang happen, and what do the cylinder and cone pictures really mean?

Answer. In this model the Big Bang is not an explosion from one privileged point in some ambient space. If $a(t) \to 0$, then every separation $a(t)\,\Delta\theta$ tends to zero. The whole closed universe shrinks everywhere at once. That is why Susskind answers the question "Where did it happen?" with "Everywhere."

The lecture makes this point concrete by imagining many galaxies spread around the circle. As we run time backward, the galaxies become more and more crowded. They are not converging toward

one special place on the circle. Rather, every distance between neighboring comoving points is shrinking simultaneously. The geometry itself is collapsing.

The second part of the answer concerns curvature. A cylinder drawn in three dimensions looks curved, but in the one-space, one-time toy model of the lecture it is intrinsically flat: one can cut it open and lay it out on a plane without stretching it. That is the case $a = \text{const}$. A cone corresponds to $a \propto t$. A cone is also flat except at its tip, where there is a conical singularity. So the tip marks a special spacetime event, but the ordinary part of the cone is flat in exactly the same intrinsic sense.

The lesson is that one must not read curvature naively from an embedding diagram. The cylinder, cone, and vase pictures are pedagogical devices for visualizing the time dependence of the circle universe. The intrinsic spacetime geometry is encoded by the metric, and the Big Bang, in this closed model, is an everywhere-event in that geometry.

1.8 Summary

The lecture unfolds a single idea through increasingly richer settings. We begin with comoving labels on a line and discover that physical distance is $d = a\,\Delta x$. Differentiating at fixed label separation gives Hubble's law, $\dot{d} = (\dot{a}/a)d$. That same scale factor then reappears in the spatial metric, $ds^2 = a^2 dx^2$, and finally in the spacetime metric, $d\tau^2 = dt^2 - a^2 dx^2$, or in three dimensions $d\tau^2 = dt^2 - a^2(dx^2 + dy^2 + dz^2)$.

Homogeneity and isotropy justify the extreme simplicity of the metric on large scales, though only as empirical and scale-dependent facts within the observable universe. The closing circle model then shows how a universe may be closed and bounded without boundary, while still obeying a local Hubble law. The payoff is conceptual as much as mathematical: cosmic expansion is not matter flying out from one point into pre-existing space. It is the geometry of space itself changing its scale everywhere at once.

Chapter 2

Cosmology Lecture 2: From the Expanding Metric to Friedmann's Equation

We were talking last time about the metric of space. This lecture picks up exactly there and asks the next question: what equation tells the scale factor how to change with time? Susskind announces the plan immediately. First a little geometry, then dynamics; and the dynamics, at least for now, will be derived not from general relativity but from Newton's equations. That choice gives the lecture its shape. We begin with an expanding comoving grid, pass through a deliberately simple projectile problem, and then watch that one-body Newtonian problem turn into Friedmann's equation. The selected board frames are kept below as local mathematical evidence, following Leonard Susskind's lecture with supporting curation from LazyingArt LLC and Video2Book.

2.1 From Homogeneity to the Expanding Metric

The standing assumptions are homogeneity and isotropy. Under those assumptions Hubble's law is not something tacked on from observation after the fact; it is the natural kinematics of a uniformly expanding universe. At a fixed time,

$$v = H(t)\, d. \tag{2.1}$$

The proportionality is linear in the distance, and the important qualification comes immediately: the so-called Hubble constant is constant in space, not in time. Homogeneity means it does not vary from place to place. It does *not* mean it remains unchanged as the universe evolves.

Before writing any metric, Susskind pauses over the question of measuring rods. If space expands, why do meter sticks not expand? The lecture's answer is concrete. The forces binding ordinary matter are enormously stronger than the tiny effect of cosmological expansion on atomic scales. One may imagine adding a minuscule extra repulsion between atoms; it would shift an equilibrium separation a little, but it would not tear apart a tightly bound rod. The cosmological expansion is therefore seen not in bound objects but in the growing separation of sufficiently distant, sufficiently unbound systems such as galaxies.

This is the reason for comoving coordinates. We assign coordinates that move with the Hubble flow. A galaxy with no extra peculiar motion then keeps the same coordinate value for all time. What

changes is the relation between coordinate separation and physical distance measured in meters. That changing conversion factor is the scale factor $a(t)$.

For flat space the spatial line element is

$$ds^2 = a(t)^2 \left(dx^2 + dy^2 + dz^2\right). \tag{2.2}$$

Here x, y, z are comoving coordinates. If two galaxies have a fixed coordinate separation Δx along one axis, then their physical distance is

$$d(t) = a(t)\,\Delta x. \tag{2.3}$$

Differentiating with respect to time gives

$$v = \dot{d} = \dot{a}\,\Delta x = \frac{\dot{a}}{a}\,d. \tag{2.4}$$

So once the comoving metric is in hand, the Hubble parameter becomes

$$H(t) = \frac{\dot{a}}{a}. \tag{2.5}$$

That is already the first bridge from geometry to dynamics.

2.2 Flat Spatial Slices Inside Curved Spacetime

Now we shift from the metric of space to the metric of spacetime. The lecture insists on an important point: three-dimensional space may be flat while spacetime is nevertheless curved. If the geometry of space changes with time, then spacetime is no longer Minkowski even if each spatial slice remains flat. Susskind's image is a flat rubber sheet being stretched by distant giants. The sheet stays flat as a sheet, but the four-dimensional history is curved.

Starting from the special-relativistic form $d\tau^2 = dt^2 - ds^2$, we upgrade the spatial metric to

$$d\tau^2 = dt^2 - a(t)^2 \left(dx^2 + dy^2 + dz^2\right). \tag{2.6}$$

The coefficients do not depend on position in space; they depend only on time. That is the metric expression of homogeneity. In matrix language, completing the spoken board description in the standard way, one writes

$$g_{\mu\nu} = \mathrm{diag}\Big(1, -a(t)^2, -a(t)^2, -a(t)^2\Big). \tag{2.7}$$

The off-diagonal terms vanish. There are no $dx\,dy$ or $dx\,dt$ terms. The metric is almost as simple as that of flat spacetime, except that the conversion between spatial separation and time is now time-dependent.

The picture is not decoration. It carries the lecture's intuition. Alice and Bob remain one coordinate unit apart, but one coordinate unit comes to represent more and more physical distance as time goes on. If $a(t)$ were driven to zero, then every physical separation would be driven to zero with it. In the lecture this is said in plain language: everything gets squashed on top of everything else, and the density becomes infinite. If $a(t)$ grows, the universe becomes correspondingly more diffuse.

There is another subtle point here. This metric has the stated form only in the frame moving with the galaxies. Rotations leave the spatial part unchanged, but a Lorentz boost does not preserve the same isotropic appearance. Matter itself picks out the preferred comoving frame.

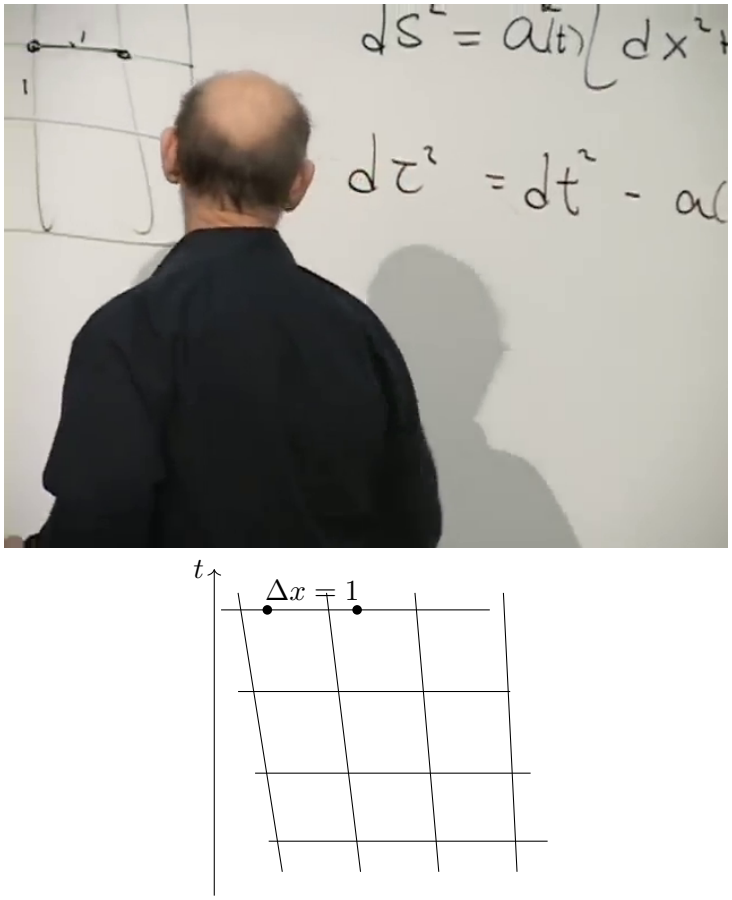


Figure 2.1: Spacetime interval and expanding-grid sketch. The lecture frame is retained as board evidence; the redraw beneath it makes the comoving grid legible as a geometric cartoon of fixed coordinate separation and increasing physical distance.

2.2.1 Question & Answer

Whose time is t in $a(t)$? It is the time measured by clocks moving with the galaxies. The lecture even says: imagine a cosmic clock factory making identical clocks, one carried by each galaxy. The coordinate time t is the time those comoving clocks measure.

This is also immediate from the metric. Along a galaxy worldline in the comoving frame we have

$$dx = dy = dz = 0, \tag{2.8}$$

so the interval reduces to

$$d\tau^2 = dt^2, \tag{2.9}$$

hence

$$dt = d\tau. \tag{2.10}$$

So the time appearing in $a(t)$ is cosmic proper time along the flow lines of the galaxies.

2.3 Newtonian Warm-Up: A Projectile in a Gravitational Field

With the geometry fixed, Susskind makes the lecture's second major pivot. Rather than deriving the dynamics of $a(t)$ from general relativity, he cools the problem down first and goes back to a familiar Newtonian one-body system. Before doing cosmology, let us understand a particle ejected radially outward from a heavy gravitating mass.

Take a heavy mass M and a much smaller mass m. Give the small mass an outward impulse and then let it move freely under gravity. Angular complications are not the point here, so the motion is taken to be radial. What matters for the cosmological analogy is energy conservation.

The kinetic energy is

$$\frac{1}{2}mv^2. \tag{2.11}$$

The gravitational potential energy has the product of the masses, Newton's constant, and one power of the distance in the denominator:

$$-\frac{GMm}{d}. \tag{2.12}$$

On the board the factors are written in the order mMG/D; in the cleaned equations we restore the conventional form GMm/d, but the board order is kept visible in the screenshots below. The conserved total energy is then

$$E = \frac{1}{2}mv^2 - \frac{GMm}{d}. \tag{2.13}$$

Here "constant" means constant in time for the particular particle under discussion. Different projectiles may of course have different values of E.

Notice how the lecture builds the formula. First the kinetic term is written. Then the potential term is added. Only after that does the negative sign become the issue. The sequence matters because later the cosmological argument will lean on the same intuition: not just on the existence of a formula, but on what the sign of the gravitational term means.

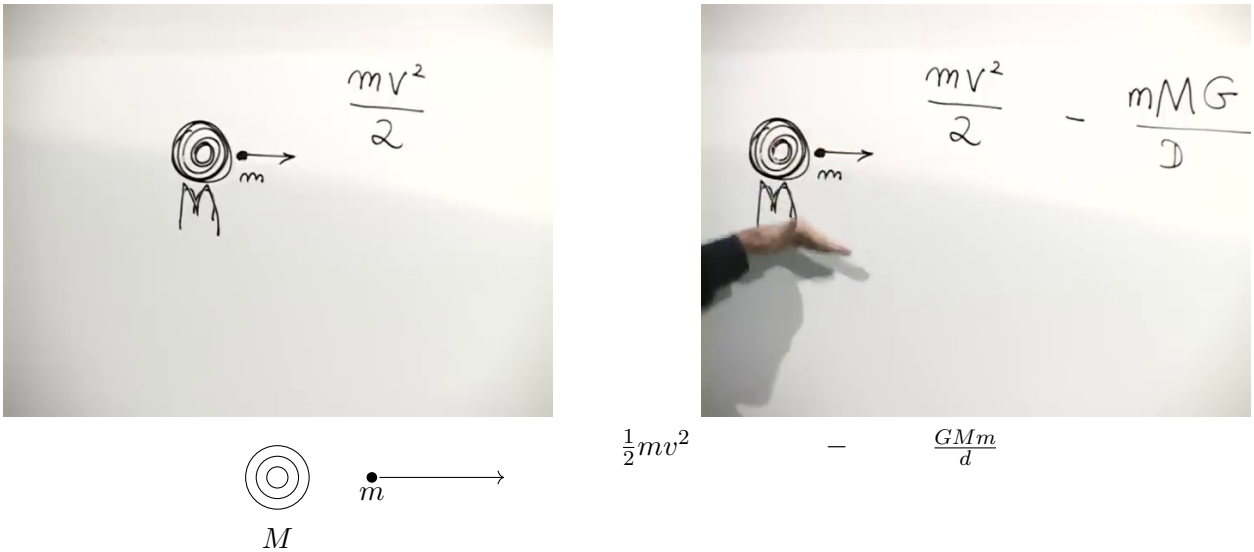


Figure 2.2: The Newtonian warm-up in the same order as the board. First the kinetic term, then the full energy balance with negative gravitational potential. The redraw regularizes the source–test-mass geometry without replacing the lecture frames.

2.3.1 Question & Answer

Why is the gravitational potential energy negative? The lecture does not answer this by abstract convention. It answers it by drawing the graphs of $1/D$ and $-1/D$. Forget the constants. The function $1/D$ decreases as D increases. The function $-1/D$ is the same curve upside down. Forces push in the direction of decreasing potential energy. Therefore an attractive force, which pulls inward toward smaller D, must correspond to the upside-down curve:

$$V(d) \propto -\frac{1}{d}. \tag{2.14}$$

Bringing the mass closer to the source creates a larger amount of negative potential energy. Moving it farther away makes the energy less negative. That is exactly what attraction means in energy language.

2.4 Escape Velocity and the Three Energy Regimes

Once the conserved energy is on the board, the lecture does not calculate right away. Instead it classifies the possible motions. There are only three cases.

- If $E < 0$, the motion is bound. The particle rises, slows, turns around, and falls back.
- If $E = 0$, the particle has exactly the escape velocity. The distance keeps increasing forever, while the speed decreases toward zero.
- If $E > 0$, the particle escapes completely. Gravity still decelerates it for a while, but not enough to prevent a nonzero asymptotic outward speed.

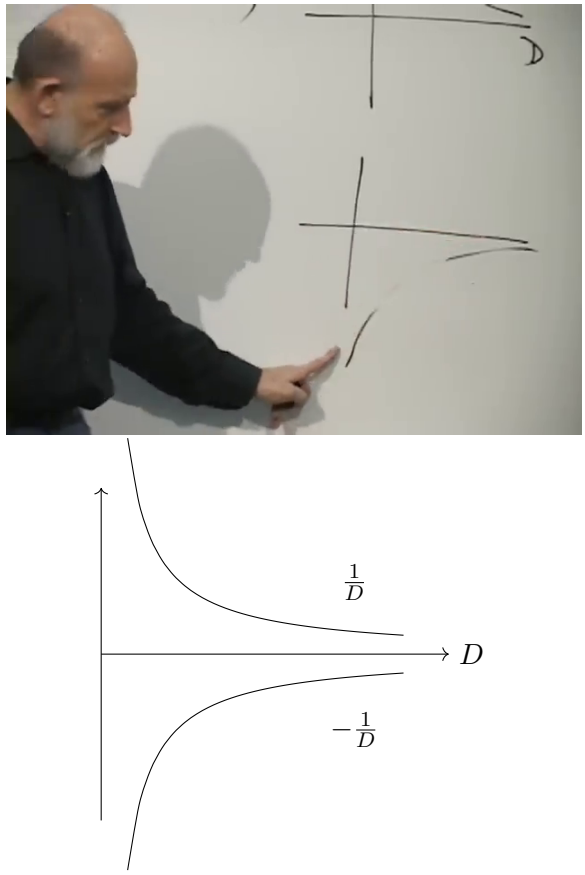


Figure 2.3: Qualitative potential-energy graphs. The lecture frame is kept because it records the actual board argument; the redraw regularizes the axes and the two comparison curves $1/D$ and $-1/D$.

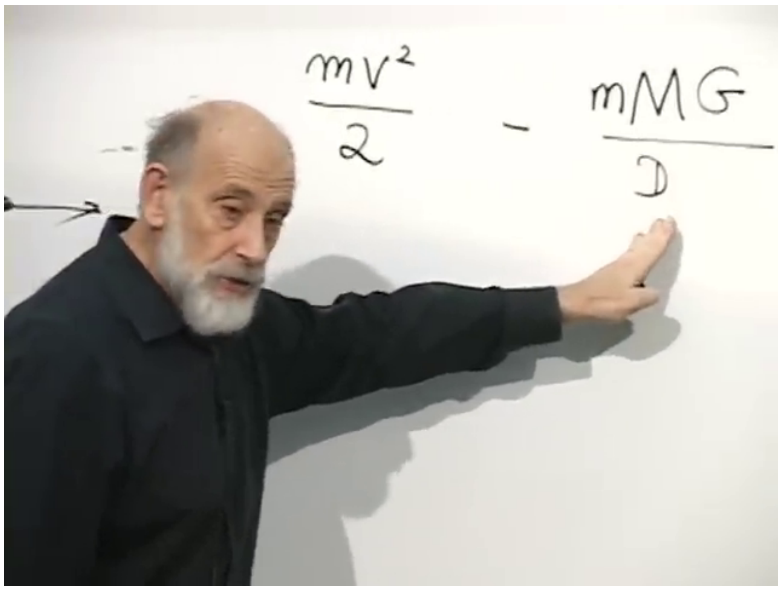


Figure 2.4: Later emphasis frame: at small enough initial outward speed, the negative potential term dominates the total energy.

The critical case $E = 0$ needs to be heard carefully. The distance is unbounded, but the speed goes to zero. It cannot actually come to rest at any finite time, because if it did, it would turn around and fall back. So the speed decreases forever without ever quite reaching zero at finite time.

For sufficiently small starting velocity the potential term dominates from the beginning, so the energy is negative:

This is the bridge to cosmology. The universe, in Newtonian language, behaves the same way. If the initial Hubble flow is large enough, galaxies escape from one another forever. If it is too small, gravity eventually wins and the universe recollapses. Between the two lies the critical case, the cosmological analog of escape velocity.

2.5 Newton's Shell Theorem and the Cosmological Reduction

The projectile analogy becomes a cosmological calculation only because of Newton's shell theorem. This is the lecture's third hinge, and Susskind takes time to make the audience trust it.

For a spherically symmetric mass distribution, the gravitational force on a particle at radius d is exactly the same as if all the mass inside radius d were concentrated at the center, while the mass outside radius d contributes no net force. The theorem is stronger than the familiar outside-a-sphere statement. A perfect spherical shell exerts no force on a particle anywhere inside it, even off-center. This is a surprising property, and in the lecture Susskind reminds us that it is special to the inverse-square law.

He even sketches a water-flow analogy: a point source produces a $1/r^2$-type flux in three dimensions, and a symmetric shell of such sources produces no net interior flow. The details are postponed, but the moral is clear. Exterior shells cancel.

Now we use that theorem in cosmology. Pick any galaxy and, for the sake of calculation, call it the

center. Because the universe is homogeneous and isotropic, this choice cannot matter. Pick a second galaxy at comoving coordinate x. Draw a sphere centered on the first galaxy and passing through the second. Then only the mass inside that sphere matters for the motion of the second galaxy.

Two further facts are essential.

First, the center is arbitrary. We could have centered the calculation on Alice, Bob, Fred, or any other galaxy, and the same equation must come out.

Second, the sphere expands with the Hubble flow. It therefore encloses the same galaxies for all time, and so its enclosed mass is constant in time even though its physical radius changes.

2.6 From Galaxy Energetics to Friedmann's Equation

We are now ready to write for a galaxy exactly the same kind of Newtonian energy equation we wrote for the projectile. After dividing by the test mass, it is convenient to write

$$\frac{v^2}{2} - \frac{GM(d)}{d} = \varepsilon(x), \tag{2.15}$$

where $\varepsilon(x)$ is constant in time for the galaxy at comoving coordinate x. It is not required to be the same for every x.

The physical distance from the chosen center to that galaxy is

$$d = a(t)x, \tag{2.16}$$

and because x is comoving,

$$v = \dot{d} = \dot{a}\,x. \tag{2.17}$$

If $\rho(t)$ is the homogeneous density, then the enclosed mass is

$$M(d) = \frac{4\pi}{3}\rho\, d^3 = \frac{4\pi}{3}\rho\, a^3 x^3. \tag{2.18}$$

Substituting these into the energy equation gives

$$\frac{(\dot{a}x)^2}{2} - \frac{G\left(\frac{4\pi}{3}\rho\, a^3 x^3\right)}{ax} = \varepsilon(x), \tag{2.19}$$

$$\frac{\dot{a}^2 x^2}{2} - \frac{4\pi G}{3}\rho\, a^2 x^2 = \varepsilon(x). \tag{2.20}$$

Here the lecture lingers on the factor of x^2, and rightly so. Both terms on the left-hand side are proportional to x^2. Everything else there is the same for every galaxy. Therefore the right-hand side must also be proportional to x^2. That is the precise meaning of the seemingly casual statement that the "constant" must scale with x^2. The farther galaxies carry larger energies, and that is exactly what one should expect since they are also moving faster.

So we write

$$\varepsilon(x) = -\frac{k}{2}\,x^2, \tag{2.21}$$

where the normalization of k is conventional. Dividing through by $x^2/2$, we obtain

$$\dot{a}^2 = \frac{8\pi G}{3}\rho\, a^2 - k. \tag{2.22}$$

Dividing once more by a^2, we arrive at the standard historical form,

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}. \tag{2.23}$$

This is Friedmann's equation in the Newtonian form emphasized in the lecture. One can also write it as

$$H(t)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}, \tag{2.24}$$

since $H = \dot{a}/a$. Susskind likes to pause here and point out the historical drama of the result: a purely Newtonian energy argument has landed on the same equation that later appears as a component of Einstein's equations.

The sign of k tracks the same three cases we already met in the escape-velocity problem. The critical case is $k = 0$. In units with $c = 1$, k is dimensionless.

2.7 Matter Density, the Critical Case, and the Age of the Universe

If we sent ourselves home at this point and tried to solve Friedmann's equation, the first objection would be immediate: ρ is not a constant. It changes as the universe expands. So the next task is to express that dependence cleanly.

Susskind does it with a unit comoving cube. Take a cube that is one unit on a side in coordinate space. The galaxies move with the comoving coordinates, so the same galaxies remain inside that cube for all time. Let the mass inside it be $m_\square$. Then the physical side length of the cube is $a(t)$, and the physical volume is $a(t)^3$. Therefore

$$\rho(t) = \frac{m_\square}{a(t)^3}. \tag{2.25}$$

Substituting this into Friedmann's equation gives

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G\, m_\square}{3a^3} - \frac{k}{a^2}. \tag{2.26}$$

Now choose the simplest case, $k = 0$, the analog of exact escape velocity. Then

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G\, m_\square}{3a^3}. \tag{2.27}$$

Before solving, the lecture points out a qualitative consequence. The right-hand side is always positive. Therefore the left-hand side cannot pass through zero, which means the model does not turn around. In the critical case the universe expands forever, but with a continually decreasing expansion rate.

Now comes the worked solution. Rather than grinding through the differential equation abstractly, Susskind tries a power law:

$$a(t) = ct^p. \tag{2.28}$$

Then

$$\dot{a} = cp\, t^{p-1}, \qquad \frac{\dot{a}}{a} = \frac{p}{t}. \tag{2.29}$$

Substituting into the equation gives

$$\frac{p^2}{t^2} = \frac{8\pi G\, m_\square}{3c^3\, t^{3p}}. \tag{2.30}$$

For this to hold for all t, the powers of t must match:

$$3p = 2. \tag{2.31}$$

So

$$p = \frac{2}{3}. \tag{2.32}$$

Then matching coefficients gives

$$\frac{4}{9} = \frac{8\pi G\, m_\square}{3c^3}, \tag{2.33}$$

hence

$$c^3 = 6\pi G\, m_\square. \tag{2.34}$$

The critical matter-dominated solution is therefore

$$a(t) = (6\pi G\, m_\square)^{1/3}\, t^{2/3}. \tag{2.35}$$

More generally one may absorb the remaining integration constant into a shift of the origin of time and write

$$a(t) \propto (t - t_0)^{2/3}. \tag{2.36}$$

This is a more interesting result than it may look at first sight. The universe does not expand linearly with time. Linear growth would be the analog of unaccelerated motion. Here the growth is slower than linear:

$$a(t) \propto t^{2/3}. \tag{2.37}$$

So the expansion slows down, but in the critical case it never turns around.

The interpretation of $t = 0$ is also part of the lecture's rhythm. If $a(t)$ really went to zero, everything would be squashed on top of everything else and the density would diverge. Susskind immediately softens the literalness of that statement: the Newtonian model cannot be trusted all the way down to infinite density. But it does tell us that the universe emerged from an extremely dense early state. That is the Big Bang limit of the model.

The Hubble parameter now becomes explicit:

$$H(t) = \frac{\dot{a}}{a} = \frac{2}{3t}. \tag{2.38}$$

So in this model

$$t = \frac{2}{3H}. \tag{2.39}$$

This is one of the lecture's payoffs. The Hubble constant is not a constant at all; it falls like $1/t$. Susskind jokes that a final examination could consist of a single question: *Is the Hubble constant a constant?* The correct answer is no.

The same formula turns a present-day measurement of H into an age estimate. In the lecture Susskind quotes an age of roughly 13 to 14 billion years and remarks that $k = 0$ is observationally very realistic, with no known significant deviation from flatness at the level of precision he quotes.

2.7.1 Question & Answer

Why does the age estimate not depend on the arbitrary fiducial box mass? The lecture's late discussion of this point is subtle because the notation becomes slippery, but the underlying idea is clean. The fiducial comoving box is conventional. If we double its side length, then the mass inside it grows by a factor of 8, while $m_\square^{1/3}$ grows by a factor of 2. The corresponding normalization of a changes by the same factor. What is physical is the density

$$\rho = \frac{m_\square}{a^3}, \tag{2.40}$$

or equivalently the invariant combination $a/m_\square^{1/3}$, not a and $m_\square$ taken separately.

The cancellation appears directly in the solution. If

$$a(t) = ct^{2/3}, \tag{2.41}$$

then

$$\dot{a} = \frac{2}{3} c\, t^{-1/3}, \tag{2.42}$$

so

$$\frac{\dot{a}}{a} = \frac{\frac{2}{3} c\, t^{-1/3}}{ct^{2/3}} = \frac{2}{3t}. \tag{2.43}$$

The constant c, and therefore the dependence on the fiducial-box mass, drops out of $H = \dot{a}/a$. That is why the age relation

$$t = \frac{2}{3H} \tag{2.44}$$

does not depend on the arbitrary normalization of the comoving box in the critical model.

This is also the right place to hear the lecture's last reinterpretation. Rearranging Friedmann's equation gives

$$\frac{\dot{a}^2}{a^2} - \frac{k}{a^2} = \frac{8\pi G}{3}\rho. \tag{2.45}$$

The left-hand side is geometric: it is built from the scale factor and, next time, from spatial curvature. The right-hand side is matter density, or in relativistic language energy density. That is exactly the sort of equation Einstein's theory ought to produce.

2.8 Summary

The lecture unfolds in a very deliberate sequence. Homogeneity and isotropy fix the kinematics of expansion. The comoving metric gives the scale factor a precise meaning. A Newtonian projectile problem provides the right intuition for conserved energy and escape velocity. Newton's shell theorem then converts the many-body universe into an effective one-body problem, and that reduction yields

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}. \tag{2.46}$$

In the critical matter-dominated case $k = 0$, the solution is

$$a(t) \propto t^{2/3}, \qquad H(t) = \frac{2}{3t}. \tag{2.47}$$

So the Newtonian model already gives a serious cosmological prediction: a slowing but never reversing critical expansion, together with an age estimate from the present Hubble parameter. The next lecture will reopen the equation from the relativistic side and show how curvature and Einstein's equations complete the story.

Chapter 3

Reviewing Friedmann, Adding Radiation, and Finding Dark Matter

This lecture is staged very deliberately. We begin in review mode, going back through the basic cosmological equation in the simplest possible setting, not because Newton is the last word, but because in this setting Newton and Einstein momentarily say the same thing. Then we change only one ingredient at a time: first the interpretation of ρ, then the dominant contents of the universe, then the observational consequences of the resulting scaling laws. Only after that cosmological spine is in place do we turn to dark matter and use the same Newtonian instincts again, now on galactic scales. These notes follow Leonard Susskind's lecture order and rhythm, with transcript and frame curation indebted to LazyingArt LLC and Video2Book.

3.1 Homogeneity, Comoving Coordinates, and the Newtonian Setup

Susskind opens by reminding us that on scales larger than clusters of galaxies, the universe looks smooth enough to be treated as homogeneous. That does not mean it is literally featureless. It means that when we average over sufficiently large distances, the jagged astronomical picture gives way to something simple enough for cosmology.

With that approximation in hand, we choose ourselves to sit at the origin of coordinates. This choice has no physical privilege attached to it; it is merely a convenient orientation. We then place a representative galaxy, or if we want to strip away all unnecessary baggage, a purely imaginary test particle moving with the Hubble flow, at comoving coordinate $x = 1$. We are at $x = 0$.

Now the scale factor is introduced in the lecture in the most concrete possible way. The physical distance to the particle at $x = 1$ is, by definition,

$$a(t). \tag{3.1}$$

A particle at $x = 2$ is then at distance $2a(t)$, and so on. In other words, $a(t)$ is not first presented as an abstract metric coefficient; it is the physical distance to a conveniently chosen comoving particle.

The corresponding Newtonian picture is just as simple. We draw a sphere centered on us, with radius equal to the particle's distance $a(t)$. The only gravitating material that matters for the

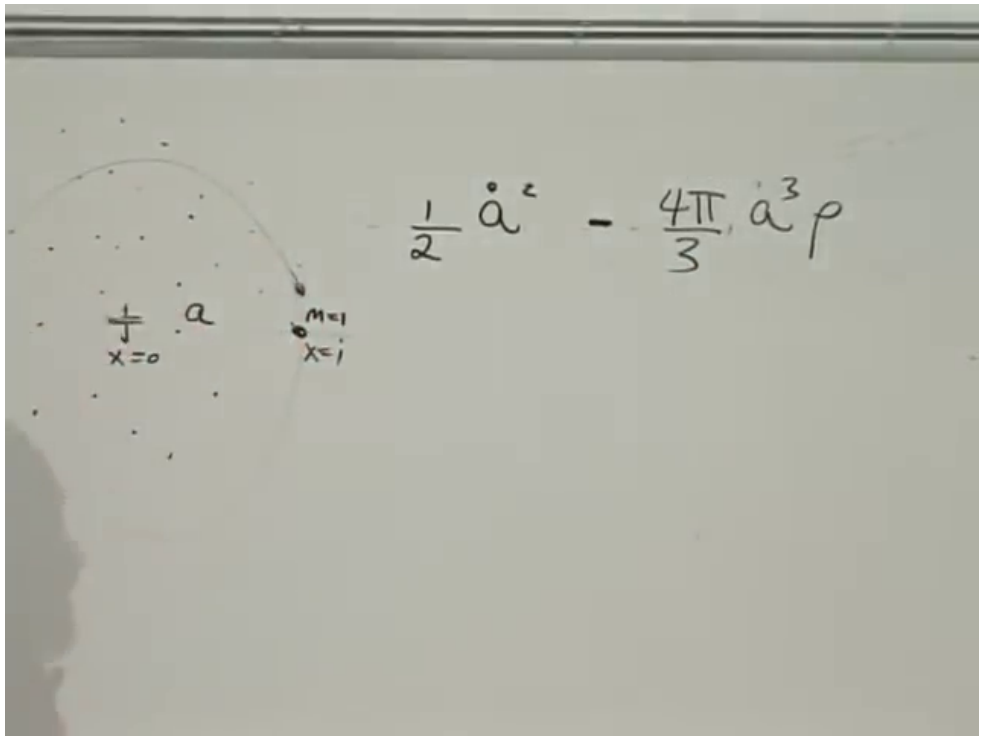


Figure 3.1: Energy-conservation term with spherical mass sketch

motion of the particle is the material inside that sphere. So the whole cosmological problem is reduced, at first pass, to the motion of one particle under the gravity of the homogeneous matter enclosed within a sphere of radius $a(t)$.

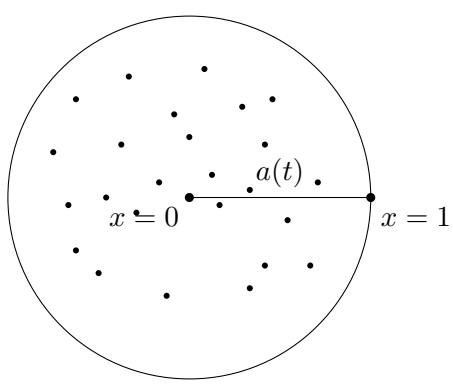


Figure 3.2: Clean reconstruction of the homogeneous sphere used in the lecture

3.2 Energy Conservation and the Friedmann Equation

At this point Susskind says, in effect: let us just go through the argument again, because it is really very simple. We assign the test particle unit mass, so that we do not have to keep writing factors

that cancel out anyway. Its kinetic energy is then

$$\frac{1}{2}\dot{a}^2. \tag{3.2}$$

The dot denotes the derivative with respect to cosmic time.

The mass enclosed inside the sphere of radius $a(t)$ is

$$M_{\text{enc}} = \frac{4\pi}{3}a^3\rho. \tag{3.3}$$

Here ρ is initially being spoken of in Newtonian language as a mass density. The gravitational potential energy of the particle is therefore

$$-\frac{GM_{\text{enc}}}{a} = -\frac{4\pi G}{3}a^2\rho. \tag{3.4}$$

The first blackboard frame only secures this term partially; the factor of G and the precise power of a are restored from the spoken derivation rather than from the chalk alone.

So the basic energy balance is

$$\frac{1}{2}\dot{a}^2 - \frac{4\pi G}{3}a^2\rho = \text{const.} \tag{3.5}$$

Newton tells us that this constant does not depend on time. In the lecture Susskind first names the constant $-k$, then immediately remarks that the standard normalization really introduces k only after multiplying by 2. The clean way to write the result is therefore

$$\dot{a}^2 = \frac{8\pi G}{3}a^2\rho - k, \tag{3.6}$$

and then, dividing by a^2,

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}. \tag{3.7}$$

This is the Friedmann equation in the form that will drive the entire rest of the lecture.

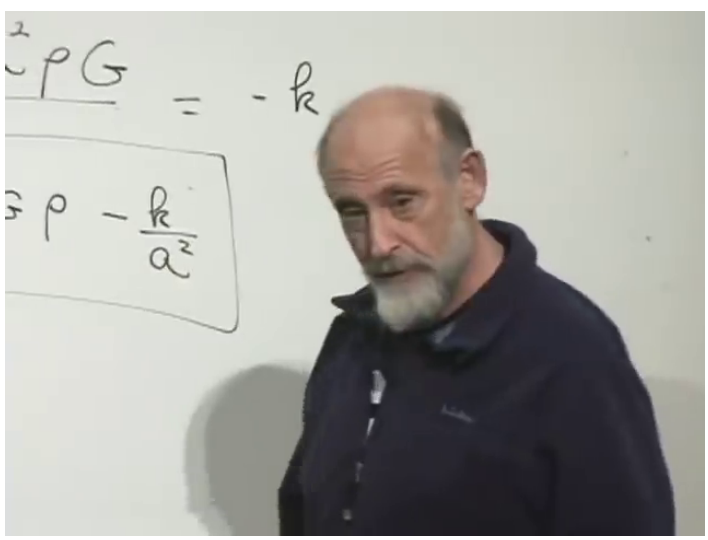


Figure 3.3: Boxed curvature term in the Friedmann equation

The board frame above is useful in a narrow but important way: it shows that the curvature term $-k/a^2$ was explicitly boxed and emphasized. Susskind also notes that k is empirically very close to zero. It is related to the spatial curvature, but observationally it is so small that, at the level of this lecture, it is a very good approximation simply to ignore it.

This is also the point at which he briefly pivots back toward Einstein. The aim is not to rederive general relativity, but to orient us. In schematic form the time-time Einstein equation looks like

$$R_{00} - \frac{1}{2} g_{00} R \propto T_{00}. \tag{3.8}$$

If we insert an expanding homogeneous metric, the geometric side produces the combination of $\dot{a}/a$ and k/a^2, while the matter side produces the energy density. So this Newtonian-looking equation is, in fact, the 00-component of Einstein's equation for a homogeneous universe. In Susskind's phrase, we have not abandoned Albert for Isaac; at this stage Isaac and Albert are saying the same thing.

That remark also changes the meaning of ρ. From now on it is better to call ρ the energy density. For nonrelativistic matter at rest in the comoving frame, energy density and rest-mass density are the same thing once we set $c = 1$. But the distinction will matter as soon as radiation enters.

3.3 Matter-Dominated Cosmology

The next move in the lecture is: now let us assume a form for ρ. Since our coordinates move with the material, the matter is at rest in the coordinate frame we are using. Under those conditions, ρ is simply the rest-energy density.

Take a cube one coordinate unit on a side. Its physical side lengths are all $a(t)$, so its physical volume is $a^3(t)$. Suppose the total mass-energy inside that comoving cube is M, and suppose it stays there: matter moves with the flow, but it does not cross the boundary of the comoving box. Then

$$\rho_m = \frac{M}{a^3}. \tag{3.9}$$

This is the matter-dominated case.

Susskind is careful about what that phrase means. Matter-dominated does not mean merely that there are particles present. It means the dominant contribution to the energy density comes from particles whose energies are essentially their rest energies in the comoving frame. In that limit the pressure is approximately zero. In the language of the lecture, pressure comes from particles crossing the boundaries of the box; here they do not. Strictly speaking, our present universe is not matter-dominated because dark energy now dominates the cosmic energy budget, but the ordinary matter sector and a long stretch of cosmological history are accurately described this way.

With $k \approx 0$, the Friedmann equation becomes

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} \frac{M}{a^3}. \tag{3.10}$$

Now comes the "honorable" method of guessing a solution. We try a power law,

$$a(t) = Ct^p. \tag{3.11}$$

Then

$$\dot{a} = Cp\,t^{p-1}, \tag{3.12}$$

$$\frac{\dot{a}}{a} = \frac{p}{t}, \tag{3.13}$$

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{p^2}{t^2}. \tag{3.14}$$

Substituting into the matter-dominated equation gives

$$\frac{p^2}{t^2} = \frac{8\pi G}{3}\frac{M}{C^3 t^{3p}}. \tag{3.15}$$

The powers of t must match, so

$$3p = 2, \qquad \text{hence} \qquad p = \frac{2}{3}. \tag{3.16}$$

Therefore

$$a(t) \propto t^{2/3}. \tag{3.17}$$

If we also match the coefficients, we recover Susskind's bookkeeping constant,

$$C^3 = 6\pi G M. \tag{3.18}$$

This is the first concrete cosmological law of the lecture: in a matter-dominated universe the scale factor grows not linearly in time, but like the two-thirds power of time. In the lecture he also sketches the curve and stresses that it rises more slowly than a straight line. Historically, this law describes a long intermediate era, roughly from about 10^5 years after the Big Bang until a few billion years later.

3.4 Radiation-Dominated Cosmology

Now we run the universe backward. Earlier than the matter-dominated era, the lecture says, the energy density was dominated not by matter but by radiation. The reason is not mysterious once we phrase it correctly: the universe contains an enormous number of photons, roughly 10^9 photons for every proton, and when we compress the universe those photons become much more energetic.

Susskind pauses to say something important. A photon has no rest mass, but that does not mean it has no energy. We are not talking about rest energy for photons. If one tried to bring a photon to rest, there would be no particle left over. But a moving photon certainly has energy, and it is that energy that matters cosmologically.

So again we take a comoving unit box. Photons move through the box, but on average as many enter as leave, so the average number inside stays fixed. The energy of a single photon is

$$E_\gamma = h\nu = \frac{hc}{\lambda}. \tag{3.19}$$

The key physical assumption, which the lecture treats as well justified by wave dynamics, is that the wavelength stretches with the expansion:

$$\lambda \propto a. \tag{3.20}$$

Therefore

$$E_\gamma \propto a^{-1}. \tag{3.21}$$

If the average number of photons in the comoving box stays fixed, then the total energy in the box scales the same way:

$$E_{\text{box}} \propto a^{-1}. \tag{3.22}$$

Dividing by the physical volume a^3, we get

$$\rho_r \propto a^{-4}. \tag{3.23}$$

This extra factor of a^{-1} is the new ingredient. Matter dilutes only because the universe gets bigger. Radiation dilutes for that reason and also because each individual photon is redshifted.

It is useful to collect the scaling chain in one place:

$$E_\gamma = \frac{hc}{\lambda}, \tag{3.24}$$

$$\lambda \propto a, \tag{3.25}$$

$$E_\gamma \propto a^{-1}, \tag{3.26}$$

$$E_{\text{box}} \propto a^{-1}, \tag{3.27}$$

$$\rho_r = \frac{E_{\text{box}}}{a^3} \propto a^{-4}. \tag{3.28}$$

The radiation-dominated Friedmann equation is therefore

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\frac{M_r}{a^4}, \tag{3.29}$$

where M_r is simply a constant setting the scale of the radiation energy density. In the lecture the same spoken symbol gets reused here, but it is no longer the matter constant from the previous section; it is cleaner to rename it.

We solve in exactly the same way as before. Try

$$a(t) = Ct^p. \tag{3.30}$$

Then the left-hand side scales as t^{-2}, while the right-hand side scales as t^{-4p}. Matching powers gives

$$4p = 2, \qquad \text{so} \qquad p = \frac{1}{2}. \tag{3.31}$$

Thus the radiation-dominated universe expands as

$$a(t) \propto t^{1/2}. \tag{3.32}$$

And if we keep the coefficient,

$$C^4 = \frac{32\pi G}{3} M_r. \tag{3.33}$$

3.4.1 Question & Answer

Question. Photons have no rest mass. Why is there still a factor of G in the radiation equation?

Answer. Because what gravitates is energy, not rest mass by itself. When particles move slowly, the distinction is easy to miss because their energy is mostly just rest energy. But photons are the example that forces the point. They have no rest mass and yet they carry energy, momentum, and pressure. Susskind answers this in the lecture by going back to Einstein's old box-of-photons argument: a box full of radiation has more inertia than the same box without the radiation, so by the equivalence principle it also has more weight. In cosmology that statement is summarized by putting energy density on the right-hand side of the Friedmann equation.

Question. If each photon loses energy as the universe expands, what happened to conservation of energy?

Answer. The lecture's answer is not that conservation failed, but that one has to keep the whole Friedmann bookkeeping in view. If we restore the full energy balance,

$$\frac{1}{2}\dot{a}^2 - \frac{4\pi G}{3}a^2\rho = \text{const}, \tag{3.34}$$

then the redshifting of the photon bath is only one part of the story. There is also the kinetic energy of the expansion and the gravitational part of the energy balance. The energy stored in radiation decreases, but the full cosmological energy accounting is still encoded in the same conserved equation.

3.5 Matter-Radiation Equality and the Change of Expansion Law

Having extracted the two scaling laws, Susskind stops computing and turns to the board. This is an important shift in rhythm. The issue is no longer how to solve each toy model separately, but how the real universe passes from one regime to the other.

For matter we have

$$\rho_m \propto a^{-3}, \tag{3.35}$$

and for radiation

$$\rho_r \propto a^{-4}. \tag{3.36}$$

So when a is large, matter dominates; when a is small, radiation dominates. The lecture's point is that the early universe must therefore have been radiation-dominated.

The frame above preserves the blackboard moment at which he begins this plot. The completed picture shows exactly what the algebra already told us: radiation lies above matter at small a, matter lies above radiation at large a, and the two cross at an equality scale factor.

In time language, the two limiting laws are

$$a(t) \sim t^{1/2} \qquad \text{early}, \qquad a(t) \sim t^{2/3} \qquad \text{later}. \tag{3.37}$$

The lecture-level estimate is that matter-radiation equality happened after about 10^4 years, perhaps a bit more.

Susskind then draws the complementary plot, this time of scale factor versus time, to make the same point dynamically rather than statically.

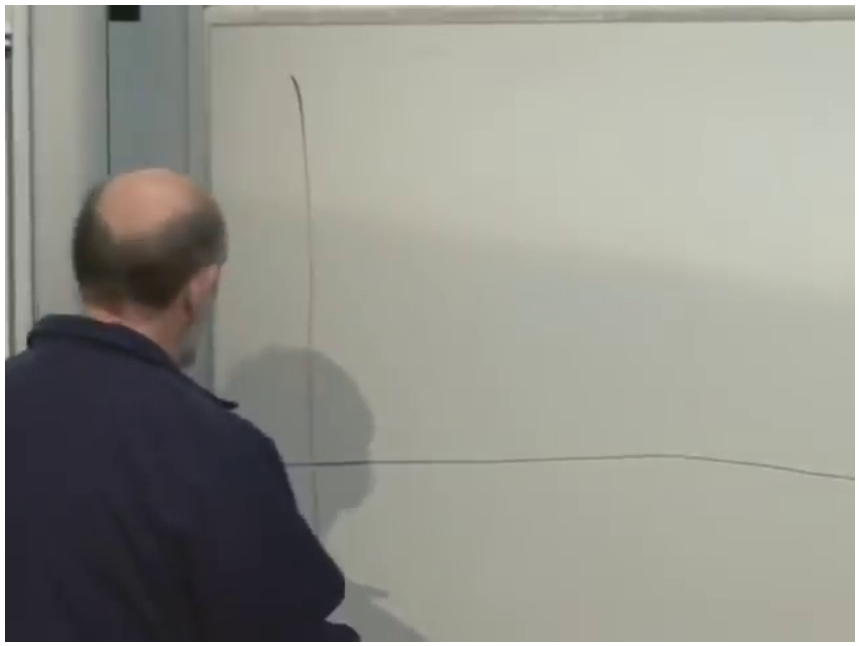


Figure 3.4: Axes for the energy-density plot

3.5.1 Question & Answer

Question. Does the power p jump from 1/2 to 2/3?

Answer. No. The lecture makes a point of correcting precisely that misconception. The exponents 1/2 and 2/3 describe the two clean limiting eras. The real universe interpolates continuously between them over a relatively narrow transition region. The effective law changes, but it does not jump discontinuously.

3.6 Temperature, Ionization, and the Surface of Last Scattering

Before estimating the crossover in more detail, Susskind suddenly says: before we talk about that, let us talk about temperature. This is a characteristic move in the lecture. The algebra has given us the scaling laws; now he wants to translate them into the thermal history of the universe.

Because the energy of each photon scales like $1/a$, the temperature of the radiation scales the same way, up to an unimportant numerical factor:

$$T \propto a^{-1}. \tag{3.38}$$

If a doubles, the radiation temperature is cut in half. Running the universe backward makes the photons hotter.

The lecture then uses a rough but effective classroom estimate. Take the present cosmic radiation temperature to be about 3 K, and take the ionizing temperature to be about 3000 K, the rough surface-of-the-sun scale repeatedly invoked in the lecture. Then

$$\frac{a_{\text{today}}}{a_{\text{ion}}} \sim \frac{3000\,\text{K}}{3\,\text{K}} \sim 10^3. \tag{3.39}$$

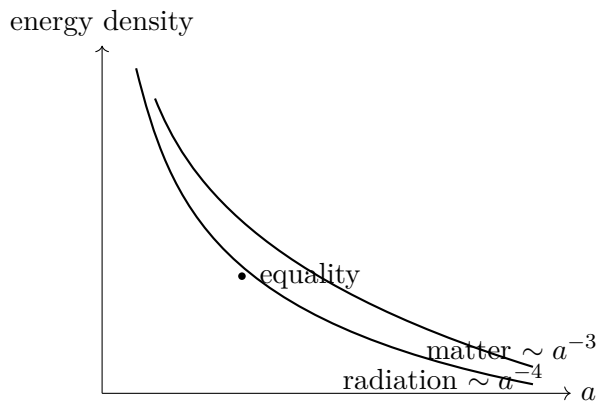


Figure 3.5: Transcript-derived reconstruction of the energy-density comparison drawn in the lecture

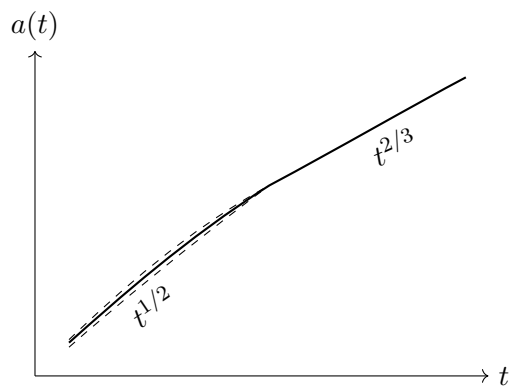


Figure 3.6: Transcript-derived sketch of the smooth transition from radiation-dominated to matter-dominated expansion

From recombination to the present the universe is well approximated as matter-dominated, so $a \propto t^{2/3}$. Therefore

$$\left(\frac{t_{\text{today}}}{t_{\text{ion}}}\right)^{2/3} \sim 10^3, \tag{3.40}$$

$$\frac{t_{\text{today}}}{t_{\text{ion}}} \sim 1000^{3/2} \sim 3 \times 10^4. \tag{3.41}$$

If we now use the lecture's order-of-magnitude value $t_{\text{today}} \sim 10^{10}$ years, we get

$$t_{\text{ion}} \sim 3 \times 10^5 \text{ years}. \tag{3.42}$$

This is not meant as a modern precision cosmological parameter. It is a lecture estimate, and it is good enough to locate the key physical event.

What is the event? Earlier than this, the universe was hot enough to ionize hydrogen and helium. It was a plasma. In a plasma, electromagnetic waves drive electric charges back and forth; in the lecture's language, the medium behaves like a conductor, and conductors are opaque. That is why the surface of the sun is opaque, and it is why the early universe was opaque. Light could not travel very far without scattering.

As the universe cooled, electrons recombined with nuclei to make neutral atoms. Once the plasma turned into ordinary neutral gas, the universe became transparent enough for photons to travel essentially freely. The transition is not infinitely sharp. Susskind stresses that it is a fuzzy place, or equivalently a fuzzy time interval, not a mathematical surface with zero thickness. But it is sharp enough cosmologically that we give it a name: the surface of last scattering.

Now the observational picture becomes vivid. An astronomer looking outward is looking backward in time. Nearby galaxies show us the relatively recent past. Farther galaxies show us earlier times. Eventually every line of sight reaches back to the epoch when the universe was still ionized and opaque. Beyond that, electromagnetic radiation cannot give us a clear view. So the cosmic microwave background is literally the radiation arriving from that last-scattering shell.

The lecture pushes the idea one step further. A neutrino telescope, if one could build a sufficiently sensitive one, would see deeper into the past, because neutrinos decouple earlier. A graviton telescope would see earlier still. But for ordinary electromagnetic astronomy the last-scattering surface is the horizon.

A final clarification comes from the question period. The temperature tracks the energy of a typical photon, but the full radiation energy density grows faster because the photon number density also changes:

$$n_\gamma \propto T^3, \qquad \rho_\gamma \propto T^4. \tag{3.43}$$

The first relation counts photons per unit volume; the second combines photon number density with energy per photon.

3.6.1 Question & Answer

Question. If the early universe was as hot as the surface of the sun in every direction, why do we not look up and see a blinding sky?

Answer. Because the last-scattering surface is not sitting still relative to us. It is receding, and the photons that left it have been redshifted on the way in. In the lecture's rough estimate, they have been redshifted by about a factor of 10^3, exactly the factor that turns a 3000 K radiation field into a 3 K microwave background. So we are indeed seeing that hot surface, but after its photons have been stretched into microwaves. Without that redshift, as Susskind says, the sky would be bright enough to burn our eyeballs.

3.7 Dark Matter from Galactic Dynamics

Only now does the lecture return to the promise made at the very beginning: dark matter. The change of scale is explicit. Dark matter was not discovered by studying cosmology in the very large-scale sense. It was discovered by astronomers looking at galaxies and measuring their rotation curves.

So Susskind goes back, once again, to the simplest Newtonian mechanics. Consider a light object of mass m orbiting a central mass M at radius r. The gravitational force is

$$\frac{GMm}{r^2}, \tag{3.44}$$

and for circular motion the centripetal acceleration is v^2/r. Newton's law gives

$$m\frac{v^2}{r} = \frac{GMm}{r^2}. \tag{3.45}$$

Cancel the small mass m, multiply through by r, and we get

$$v(r) = \sqrt{\frac{GM}{r}}. \tag{3.46}$$

Thus if the gravitating mass is concentrated near the center, the orbital speed falls like $r^{-1/2}$. This is the Keplerian law familiar from the solar system.

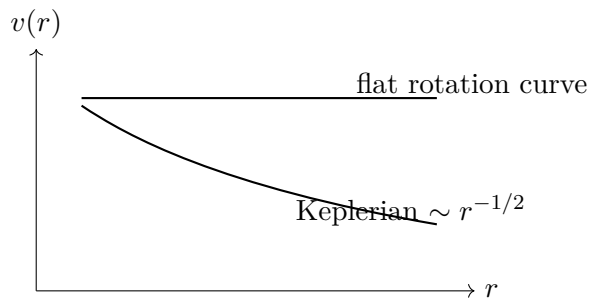


Figure 3.7: Transcript-derived comparison between Keplerian falloff and the approximately flat rotation curves seen in large galaxies

But the measured galactic rotation curves do not fall that way. If one assumes that mass follows light, so that most of the mass sits where most of the luminous matter sits, near the galactic core, then the outer stars should move more slowly. They do not. Their orbital speeds remain roughly constant out to large radii.

So the lecture changes the question. Perhaps mass does not follow light. Let $M(r)$ be the total mass enclosed inside radius r. Then the orbital relation becomes

$$\frac{v^2}{r} = \frac{GM(r)}{r^2}. \tag{3.47}$$

If $v(r)$ is approximately constant, then

$$M(r) \propto r. \tag{3.48}$$

That is already a powerful conclusion: the total gravitating mass keeps growing with radius even where the visible light does not.

Assume now, as Susskind does for simplicity, that the hidden matter distribution is roughly spherical. Then

$$M(R) = \int_0^R 4\pi r^2 \rho(r)\, dr. \tag{3.49}$$

If $M(R) \propto R$, differentiating with respect to R gives

$$4\pi R^2 \rho(R) \propto 1, \tag{3.50}$$

and hence

$$\rho(r) \propto \frac{1}{r^2}. \tag{3.51}$$

So the lecture arrives at a simple mathematical picture: a visible galaxy sits inside a much larger dark halo whose density decreases gradually rather than collapsing sharply with the light distribution.

At the numerical level Susskind plays loosely between factors of 7 and 10, and he is explicit that he is speaking heuristically. The stable statement is that the total gravitating mass of a galaxy is of order several to ten times larger than what one would infer from luminous matter alone.

From here the lecture shifts from derivation to diagnosis. Whatever this material is, it does not shine, and it evidently does not dissipate the way ordinary matter does. Ordinary gas clouds collide, radiate, and keep shrinking. This stuff stays in a large halo.

3.7.1 Question & Answer

Question. If the halo is so massive, why does it not dissipate and collapse together with the luminous matter?

Answer. Because collapse requires dissipation. Gravity by itself can pull a cloud inward, but to settle into a smaller structure the system must lose energy. Ordinary matter has many channels for doing that: collisions, friction-like processes, radiation, and electromagnetic interactions. Dark matter evidently does not. In the lecture Susskind emphasizes that it is very nearly collisionless and does not radiate appreciably. So the gravitational contraction converts potential energy into kinetic energy, but there is no efficient mechanism for dumping that kinetic energy away. What keeps the halo extended is the same thing that keeps the Earth from falling straight into the sun in the absence of friction: motion.

This is also where the lecture draws a careful boundary. The inferences $M(r) \propto r$ and $\rho(r) \propto r^{-2}$ are solid lecture mathematics. But Susskind does not claim a simple first-principles derivation of the detailed halo profile, or a detailed understanding of halo rotation, flattening, or intergalactic distribution. On those points he repeatedly says, quite openly, that he does not know and that large-scale simulations do much of the work.

The last observational beat of the lecture is the Bullet Cluster. When galaxy clusters collide, the hot ordinary gas interacts strongly and can be seen in X-rays, while the dominant gravitating component inferred from lensing passes through much more weakly interacting matter. The resulting separation between the lensing mass and the luminous baryonic material is exactly the sort of thing one would not expect if the whole phenomenon were merely a modification of Newton's force law tied to visible matter. In the lecture, this serves as the decisive piece of evidence that there really is dark matter.

Susskind ends on a note of bounded confidence. Very little is known in detail about what dark matter is, but nothing about it requires a radical rewriting of physics. The conservative possibility is that it consists of new, very weakly interacting particles. In that sense it is mysterious, but not in a way that forces us to throw away the basic principles that got us here.

3.8 Summary

The lecture begins with a single equation and then changes the meaning of its source term step by step. Starting from a homogeneous Newtonian sphere centered on us, we recover the Friedmann

equation,

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}, \tag{3.52}$$

and are reminded that this is also the time-time Einstein equation for an expanding homogeneous universe. For matter at rest in a comoving box, $\rho_m \propto a^{-3}$ and $a(t) \propto t^{2/3}$. For radiation, because photon wavelengths stretch with the expansion, $\rho_r \propto a^{-4}$ and $a(t) \propto t^{1/2}$.

From those two scaling laws come the next pieces of the story: radiation domination in the early universe, matter domination later, temperature scaling as $1/a$, recombination after roughly 3×10^5 years at lecture level, and the surface of last scattering as the electromagnetic horizon. Only then does the lecture narrow from cosmology to astronomy and use orbital mechanics to infer dark matter from flat galactic rotation curves. The mathematical punch line there is just as sharp: $v \approx$ const implies $M(r) \propto r$, hence a large dark halo with $\rho(r) \propto r^{-2}$. The details remain open, but the structure of the argument does not.

Chapter 4

Pressure, Equation of State, and Vacuum Energy

This lecture opens in the form of questions, not yet in the form of equations. Susskind first clears away several conceptual obstacles: the microwave background does define a preferred cosmological rest frame, superluminal recession does not violate local relativity, and a horizon is the place where expansion defeats return signals. Only after that warm-up does he deliberately reset the lecture and turn to the technical core: pressure, energy density, the parameter w, and the special role of vacuum energy. The aim here is to follow that unfolding as closely as possible, while keeping the mathematics clean and the board evidence visible. These notes are based on Leonard Susskind's lecture, with supporting curation by LazyingArt LLC and Video2Book.

4.1 Opening Questions: Preferred Frame, Recession, and the Microwave Background

The first question concerns the cosmic microwave background. If the background radiation looks isotropic only in one frame, does that not define a preferred frame of reference? Susskind's answer is yes, it does. That is not a contradiction of relativity. It is a statement about the large-scale state of the universe. There is a special cosmological frame in which the microwave background looks the same in all directions, and this is also, on average, the frame in which the local galaxies are at rest.

If we move through this bath of photons, the Doppler effect immediately breaks the isotropy. Photons coming from the direction of motion are blueshifted; photons from the opposite direction are redshifted. In that sense there is a preferred cosmological rest frame, but the local laws of physics remain perfectly relativistic.

The next question asks whether distant galaxies can recede faster than light. Here the essential distinction is between local motion and global recession. Relativity forbids one object from overtaking another right in front of us at superluminal speed. It does not forbid sufficiently distant galaxies from separating according to

$$v = Hd. \tag{4.1}$$

For large enough d, the recession speed v can exceed c. In the units of the lecture we set $c = 1$, so the horizon is the distance at which $Hd = 1$.

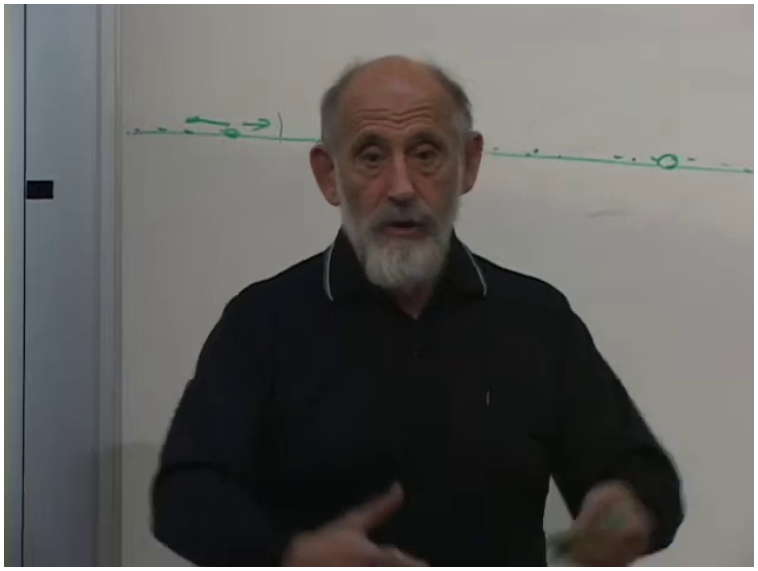


Figure 4.1: One-dimensional expansion sketch on the board. The visual evidence is sparse but important: a long line, directional arrowing, and marked points used to introduce the intuition of an expanding metric.

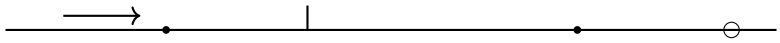


Figure 4.2: Minimal redraw of the line analogy. The lecture uses the picture qualitatively, so the redraw is kept deliberately spare.

This immediately leads to the observable universe. The cosmic horizon is the boundary beyond which light cannot get back to us. The surface of last scattering is not exactly the same thing, but it lies very close to it. Radiation from the last-scattering surface is enormously redshifted, by a factor of order 10^3, but not infinitely redshifted. For that reason the surface of last scattering sits slightly inside the horizon, while still serving in practice as the edge of what we can directly see.

4.2 Expanding Metric and the One-Dimensional Line Analogy

At this point Susskind insists that ordinary special-relativistic intuition is no longer enough. The problem is not simply one of adding velocities. In general relativity the metric itself can expand.

His favored picture is a one-dimensional line. If we stretch an ordinary material line, it thins out and changes its local properties. That is not the right analogy. The right analogy is a line with a hidden reservoir of new points. Whenever neighboring points separate too much, new points are inserted. The line grows, but its local character does not thin out.

What matters is the rule for signals. Locally, a signal always moves at the limiting speed relative to nearby points. Globally, however, the total distance it must overcome may keep increasing because more space is being inserted between source and destination. The lecture returns to this distinction again and again: local speed limits survive, even while global recession can exceed the speed of light.

Susskind also uses the same picture to explain redshift. If we draw a wave on the line, then as the line expands the wave stretches with it. In the full theory this is not merely an image. It is what

one finds by writing wave equations in an expanding geometry.

4.2.1 Question & Answer

How can a light signal fail to get back if nothing locally moves faster than light? Because the signal is not racing against a receding material object in a fixed background. It is moving through a geometry whose distance scale is itself increasing. The light ray always propagates locally at the limiting speed, but the proper distance to its destination can grow faster than the ray can eat it up. In Susskind's line picture, the signal advances from point to point while new points are being inserted in front of it. That is the cosmological horizon in a nutshell.

4.3 Reset: Friedmann Equation and the Three Components

After the opening discussion Susskind explicitly resets the lecture. He says, in effect, that this is the theoretical minimum: just enough mathematics to follow a serious cosmology argument. He dislikes jargon, but he also knows that one eventually has to understand what astronomers mean when they say that they are measuring w. So he begins again from the basic expansion equation.

The central relation is

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho, \qquad c = 1. \tag{4.2}$$

Here $a(t)$ is the scale factor, and

$$H \equiv \frac{\dot{a}}{a} \tag{4.3}$$

is the Hubble parameter. The right-hand side contains the total energy density.

The lecture then reviews the three dynamically important contributions to ρ:

- matter, meaning objects effectively at rest in the comoving frame,
- radiation, meaning photons and other effectively massless components,
- vacuum energy, written ρ_0, which does not dilute when the universe expands.

For matter and radiation the earlier results are recalled:

$$\rho_m = \frac{C_m}{a^3}, \qquad a(t) \sim t^{2/3}, \tag{4.4}$$

$$\rho_r = \frac{C_r}{a^4}, \qquad a(t) \sim t^{1/2}. \tag{4.5}$$

The third component is introduced as the odd one:

$$\rho_0 = \text{constant}. \tag{4.6}$$

Susskind does not yet derive this from first principles. He first wants to explain pressure, because pressure is what will ultimately tell us how ρ changes with a.

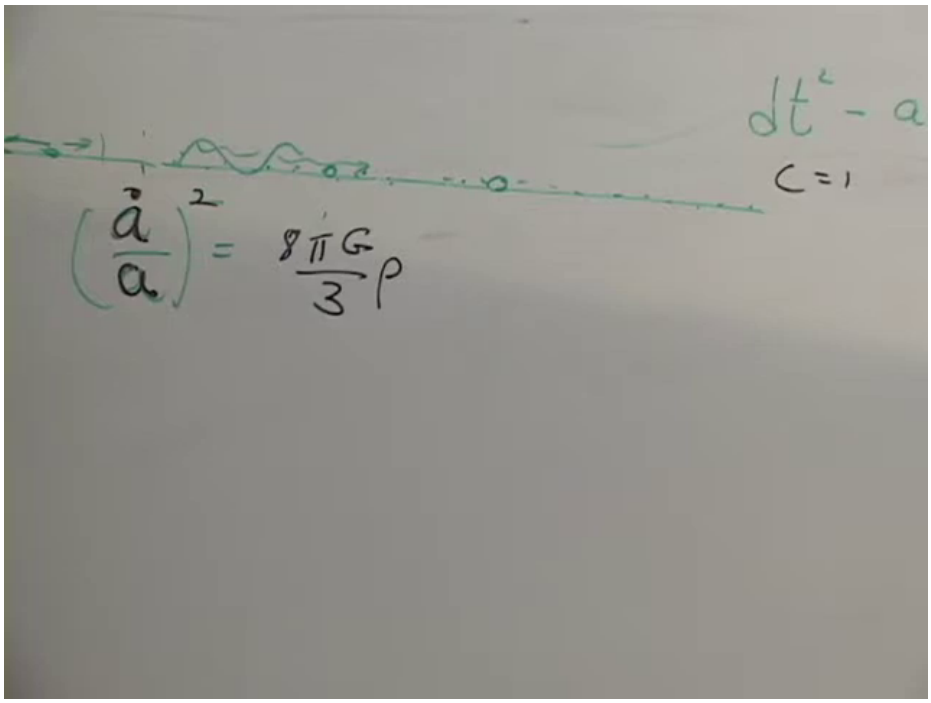


Figure 4.3: Friedmann equation with line analogy. The cropped notation on the far right only certifies that $c = 1$ was written nearby; the operative equation on the board is the Friedmann equation shown clearly here.

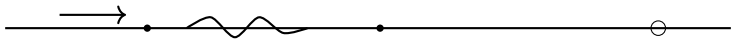


Figure 4.4: Redraw of the same line analogy, now with the small disturbance visible above the basic expansion equation.

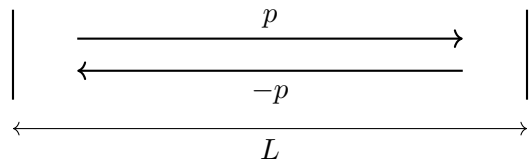


Figure 4.5: One-dimensional photon-in-a-box argument. The wall receives momentum $2p$ at each reflection, and the average force becomes energy divided by length.

4.4 Pressure from Radiation and Matter

We now come to the first real derivation of the lecture. Susskind begins with radiation, because there pressure is visible. Light bouncing off walls exerts a force, and so pressure can be computed directly from momentum transfer.

The one-dimensional box. Start with a one-dimensional box, that is, a line segment of length L, containing photons moving back and forth. In one dimension force and pressure are effectively the same thing, so this is the cleanest place to begin.

Take a single photon with momentum magnitude p. When it reflects from a wall, its momentum reverses. The change in momentum at each collision is therefore

$$\Delta p = 2p. \tag{4.7}$$

The time between successive collisions with the same wall is

$$\Delta t = \frac{2L}{c}, \tag{4.8}$$

since the photon must traverse the box and return. Hence the average force on the wall is

$$F = \frac{\Delta p}{\Delta t} = \frac{pc}{L}. \tag{4.9}$$

Now use the photon relation

$$E = pc. \tag{4.10}$$

Then

$$F = \frac{E}{L}. \tag{4.11}$$

But E/L is precisely the one-dimensional energy density. In this toy model,

$$P = \rho, \tag{4.12}$$

with the important caveat that here ρ means energy per unit length, not energy per unit volume.

With many photons the jolts are simply added and averaged. The same relation survives:

$$P = \rho \tag{4.13}$$

in one dimension.

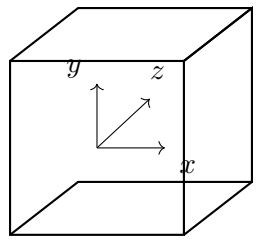


Figure 4.6: Three-dimensional schematic used to motivate the isotropic result $P_r = \rho_r/3$.

From one dimension to three. Now pass to a three-dimensional box. If all photons moved only along the x-direction, then the same reasoning would give a pressure on the walls perpendicular to x, but no pressure on the other faces. That is obviously not the correct picture of thermal radiation. Susskind deliberately makes this wrong assumption first, precisely to expose what has to be corrected.

In an isotropic radiation bath, the momentum is distributed over all directions. The pressure on any given wall depends only on the component of momentum normal to that wall. Averaged over directions, one third of the energy density contributes to each Cartesian pressure component. The result is

$$P_r = \frac{\rho_r}{3}. \tag{4.14}$$

Cold matter. The matter-dominated case is the opposite limit. Take a comoving box, meaning a box that moves with the cosmological flow. Cold matter is at rest relative to that box. The particles still contribute energy density, because each one carries rest energy, but they do not transfer momentum to the walls. Therefore

$$P_m = 0. \tag{4.15}$$

It is useful to write the board summary in clean form:

$$\text{matter:} \qquad a(t) \sim t^{2/3}, \qquad P = 0, \tag{4.16}$$

$$\text{radiation:} \qquad a(t) \sim t^{1/2}, \qquad P = \frac{\rho_r}{3}. \tag{4.17}$$

4.5 Equation of State and Density Scaling

Only now does Susskind introduce the jargon he had apologized for. The point is simple: if pressure is proportional to energy density, then the proportionality constant is a useful label for the substance filling the universe. He writes the board relation as

$$P = w\rho. \tag{4.18}$$

On the board the parameter is written as ω; in these notes we normalize to the standard w. The examples already obtained give

$$w = 0 \quad \text{for cold matter,} \qquad w = \frac{1}{3} \quad \text{for radiation.} \tag{4.19}$$

One can imagine intermediate values as well. The lecture mentions, for example, the possibility of a warm component with $0 < w < 1/3$.

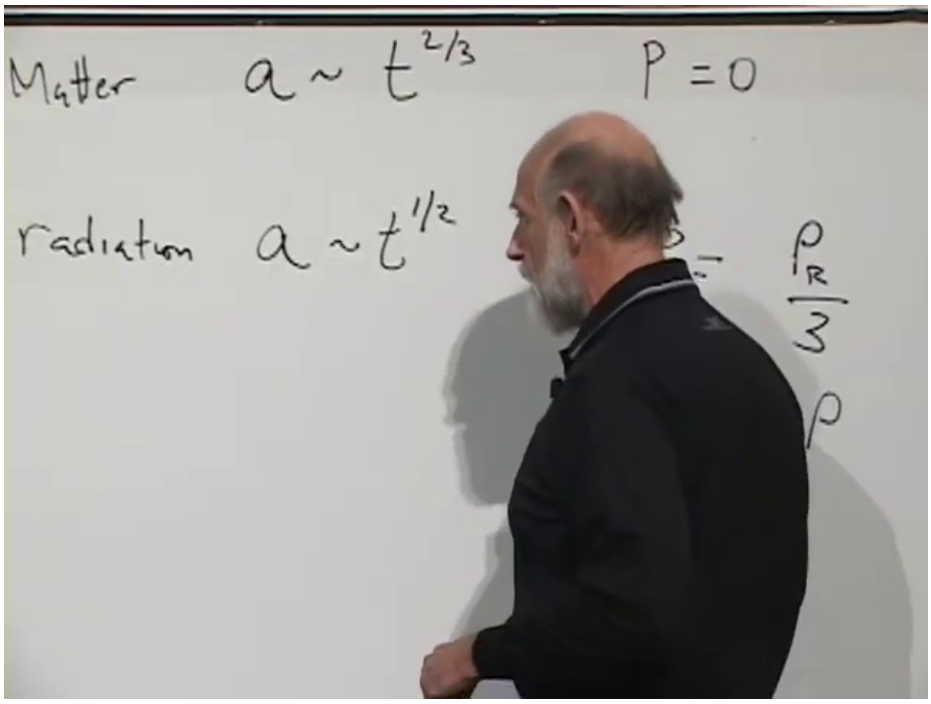


Figure 4.7: Matter and radiation equation-of-state summary on the board. The lecture uses the row-by-row layout itself as part of the argument.

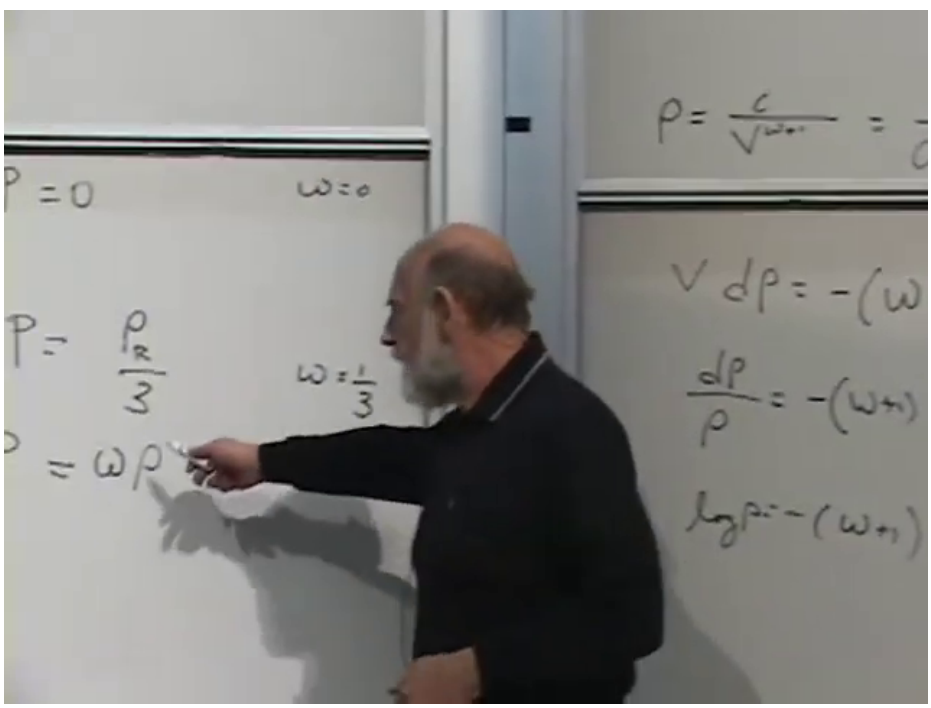


Figure 4.8: Equation-of-state parameter on the board. The left panel highlights $P = \omega\rho$, while the right panel carries the density-scaling derivation in partial form.

At this point the lecture poses the natural question. Matter has $w = 0$ and $\rho_m \propto a^{-3}$. Radiation has $w = 1/3$ and $\rho_r \propto a^{-4}$. Is this a coincidence, or is there a general relation between w and the way ρ scales?

4.5.1 Question & Answer

Why should the pressure-to-density ratio determine how the density changes with the scale factor? Because expansion does work. This is the thermodynamic heart of the lecture.

Take a box of volume V and let it expand by a small amount dV. If the contents exert pressure P, then the work done on the walls is $P\,dV$. That work must come from the internal energy of the contents, so

$$dE = -P\,dV. \tag{4.20}$$

Now write the internal energy as

$$E = \rho V. \tag{4.21}$$

Its differential is

$$dE = \rho\,dV + V\,d\rho. \tag{4.22}$$

Equating the two expressions for dE gives

$$\rho\,dV + V\,d\rho = -P\,dV. \tag{4.23}$$

Rearranging,

$$V\,d\rho = -(\rho + P)\,dV. \tag{4.24}$$

Now insert the equation of state $P = w\rho$:

$$V\,d\rho = -(1 + w)\rho\,dV. \tag{4.25}$$

Divide by ρV:

$$\frac{d\rho}{\rho} = -(1 + w)\frac{dV}{V}. \tag{4.26}$$

Integrating,

$$\ln \rho = -(1 + w)\ln V + \text{const}, \tag{4.27}$$

and therefore

$$\rho \propto V^{-(1+w)}. \tag{4.28}$$

This is the main result of the section. The rest is the subtle step that Susskind corrects on the fly during the lecture: in cosmology the comoving box has volume proportional to a^3, not to a. Thus

$$V \propto a^3, \tag{4.29}$$

and so

$$\rho \propto a^{-3(1+w)}. \tag{4.30}$$

Equivalently,

$$\rho = \frac{C}{a^{3(1+w)}} \tag{4.31}$$

for some constant C.

Now the earlier cases fall out immediately:

$$w = 0 \Longrightarrow \rho \propto a^{-3}, \tag{4.32}$$

$$w = \frac{1}{3} \Longrightarrow \rho \propto a^{-4}. \tag{4.33}$$

Matter and radiation are no longer separate facts. They are two special points on the same curve of possibilities.

Once $\rho(a)$ is known, one can feed it back into the Friedmann equation and solve for $a(t)$. Susskind does not derive the general power law in detail here; he leaves that as an exercise in trying $a \sim t^{\alpha}$. What matters for the lecture is the logic: w determines $\rho(a)$, and $\rho(a)$ determines the expansion history.

4.6 Vacuum Energy as the Special Case $w = -1$

The lecture has been steering toward one especially important value, namely

$$w = -1. \tag{4.34}$$

If we put that into the general scaling law, the exponent vanishes:

$$\rho \propto a^{-3(1+w)} = a^0. \tag{4.35}$$

Thus

$$\rho = \rho_0 = \text{const.} \tag{4.36}$$

The energy density does not dilute at all. At the same time the equation of state becomes

$$P = -\rho. \tag{4.37}$$

This is exactly the vacuum-energy case.

4.6.1 Question & Answer

How can positive energy density come with negative pressure, and why does that not make the theory inconsistent? Because negative pressure is simply tension. Positive pressure pushes on the walls of a box. Negative pressure pulls them inward. A stretched rubber band is the everyday model: it has positive energy stored in it, but it pulls its ends together. There is nothing pathological about that sign. The energy density remains positive; the pressure is negative.

Now feed ρ_0 back into the Friedmann equation:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho_0. \tag{4.38}$$

Since the right-hand side is constant, the Hubble parameter becomes a true constant,

$$H = \sqrt{\frac{8\pi G\rho_0}{3}}. \tag{4.39}$$

This is the special case in which $\dot{a}/a$ really deserves to be called a constant. The equation becomes

$$\dot{a} = Ha. \tag{4.40}$$

Its solution is exponential:

$$a(t) = a_* \exp(Ht) = a_* \exp\left(\sqrt{\frac{8\pi G\rho_0}{3}}\, t\right). \tag{4.41}$$

This is the late-time accelerated expansion. The scale factor does not merely keep increasing; its rate of increase is proportional to the scale factor itself. The characteristic timescale is H^{-1}, the e-folding time.

Susskind also remarks that the combination of ρ_0 with gravitational constants is often packaged into the cosmological constant Λ. Historically the numerical normalization is conventional and a bit untidy. For the present lecture the important physics is simpler than the terminology:

$$\Lambda \propto \rho_0. \tag{4.42}$$

Vacuum energy and cosmological constant are the same physical ingredient, up to the numerical factors attached by historical convention.

As always, the squared Friedmann equation admits two branches when one takes a square root. There is an expanding branch and a time-reversed contracting branch. The lecture, of course, follows the expanding one.

4.7 Three Eras, Observation, and the Late-Time Horizon

With the general framework in hand, Susskind steps back and looks at the whole history. The universe passes through three broad eras:

- a radiation-dominated era at small a,
- a matter-dominated era at intermediate a,
- a vacuum-dominated era at large a.

This follows immediately from the scaling laws

$$\rho_r \propto a^{-4}, \qquad \rho_m \propto a^{-3}, \qquad \rho_0 = \text{const}. \tag{4.43}$$

The time history of the scale factor reflects the same sequence:

- early on, $a(t) \sim t^{1/2}$,
- later, $a(t) \sim t^{2/3}$,
- at late times, $a(t)$ bends upward toward exponential growth.

The lecture gives the crossover times only as order-of-magnitude statements. Radiation gives way to matter very early, after roughly 10^4 years in the lecture's rough accounting, well before the universe becomes transparent. Matter gives way to vacuum much later, after several billion years, roughly halfway through cosmic history on the scale Susskind has in mind.

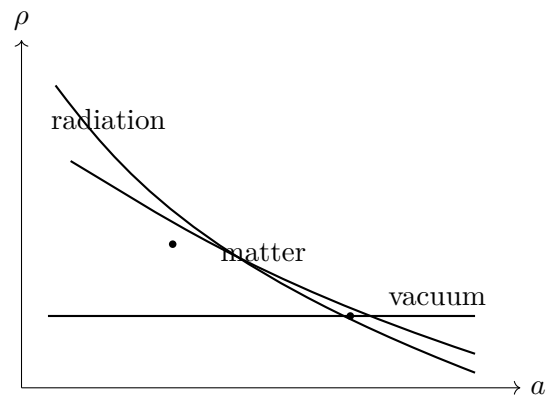


Figure 4.9: Qualitative energy-density history. Radiation dominates at very small a, matter later, and vacuum at large a.

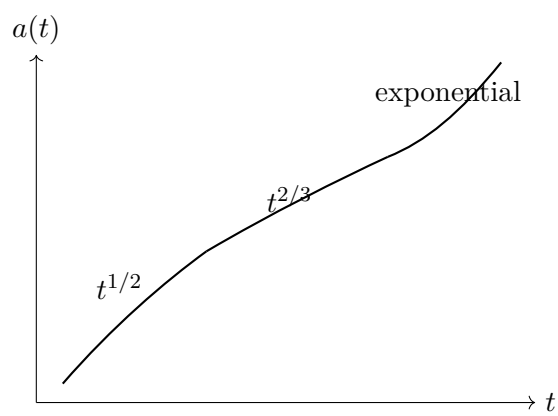


Figure 4.10: Qualitative time history of the scale factor: radiation era, matter era, then late accelerated expansion.

How do we know this sequence? We do not see the future, but we do see the past. Looking farther away means looking to earlier times. By observing distant galaxies, measuring their recession, and correlating distance with lookback time, one reconstructs the Hubble parameter as a function of time and therefore the history of $a(t)$. The radiation era is too early to be observed directly by ordinary galaxy astronomy, but the matter-dominated era and the transition to late acceleration are visible in the reconstructed expansion history.

At sufficiently late times, when vacuum energy dominates almost completely, the Hubble parameter approaches the constant value

$$H = \sqrt{\frac{8\pi G\rho_0}{3}}. \tag{4.44}$$

Using again the Hubble law $v = Hd$, the horizon distance is defined by the condition $v = 1$, that is,

$$1 = Hd_H. \tag{4.45}$$

Therefore

$$d_H = \frac{1}{H} = \frac{1}{\sqrt{8\pi G\rho_0/3}}. \tag{4.46}$$

This is the late-time cosmic horizon. Once the vacuum-dominated exponential era sets in, the horizon distance is fixed. Galaxies continue to move outward and pass through it one by one, disappearing not by abruptly vanishing, but by becoming ever more redshifted and fainter.

This is why Susskind spends time on the melancholy future of astronomy. In the remote future, only nearby gravitationally bound structures remain visible. Distant galaxies cross the horizon. The microwave background is stretched to such long wavelengths that it effectively fades away. Astronomers living then may infer that they inhabit a single isolated island of matter in an empty universe.

The lecture then briefly restores the curvature term that had been omitted in the main derivation:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{K}{a^2}. \tag{4.47}$$

Numerically this term appears to be small, but conceptually it matters. In the older cosmology, before vacuum energy was taken seriously, one imagined a matter-dominated universe with curvature of the right sign to halt the expansion and reverse it. The result was the old Big-Crunch picture: expansion, turnaround, recontraction. The mathematics is still legitimate, but it no longer appears to describe our universe.

The present-day composition, in the lecture's rough numbers, is

$$\frac{\rho_0}{\rho_{\text{total}}} \sim 0.7, \qquad \frac{\rho_m}{\rho_{\text{total}}} \sim 0.3, \tag{4.48}$$

$$\frac{\rho_{\text{lum}}}{\rho_{\text{total}}} \sim 0.05, \qquad \frac{\rho_{\text{dark}}}{\rho_{\text{total}}} \sim 0.25, \tag{4.49}$$

with radiation negligible today. Dark matter and luminous matter both scale as a^{-3}, so their mutual ratio stays roughly fixed even while the overall balance between matter, radiation, and vacuum changes from era to era.

The lecture closes by turning toward geometry. If we know the distance to the surface of last scattering, and if we have a standard lump there whose physical size can be estimated, then its observed angular size tests whether space is positively curved, negatively curved, or flat.

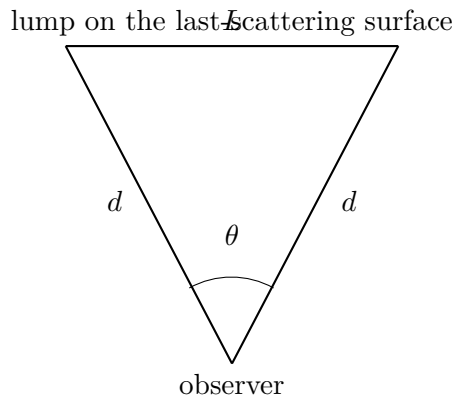


Figure 4.11: Preview of the next topic: the angular size of a standard lump on the last-scattering surface provides a test of spatial curvature.

That geometry argument belongs properly to the next lecture. Here it functions as an outlook. The work of the present lecture is already complete: pressure has been related to energy density, the equation-of-state parameter w has been introduced, and vacuum energy has emerged as the special case $w = -1$.

4.8 Summary

The lecture begins with preferred frames and horizons, but its mathematical center is the relation between pressure and energy density. Radiation gives

$$P_r = \frac{\rho_r}{3}, \tag{4.50}$$

cold matter gives

$$P_m = 0, \tag{4.51}$$

and both are unified by

$$P = w\rho. \tag{4.52}$$

Thermodynamics then shows that

$$\rho \propto a^{-3(1+w)}. \tag{4.53}$$

From that one immediately reads off the special case $w = -1$: constant vacuum energy, negative pressure, and exponential expansion. The three cosmic eras, the late-time fixed horizon, and the modern picture of an accelerating universe all follow from that single thread.